D2 Detailed Design

Verified Student Sublease Marketplace

Item	Detail
Course	CS5001 Senior Design I
Assignment	Assignment 7: Detailed System Design, Part 2 (Design D2) and Peer Review Notes
Team members	Danvanth Vanganuri, Jovan Jose Asker Fredy, Mohamed Aidja
Goal	Let UC students fill or find term-length subleases safely by limiting the marketplace to verified @mail.uc.edu students, matching listings to exact co-op dates, holding discriminatory listings for review, and giving property managers an approve/reject step before move-in.
Repository location	Design_Diagrams/D2_Detailed_Design.pdf (this document); Design_Diagrams/D2_Fig1 to D2_Fig8 (.dot sources, render with Graphviz); Design_Diagrams/Peer_Review_Notes.md
Inputs to this design	User_Stories (US-01 to US-05, UC-01, UC-02, AC-01.1 to AC-02.2); D0 High-Level Design (decision log D-01 to D-12); D1 Detailed Design (C3, C5)

1. Header, Conventions, and D2 Overview

1.1 Conventions

Every D2 figure carries the project title, the goal statement, and its own conventions block drawn on the figure. The table below collects them. Figures 2 and 3 keep D1's data-model notation, with one addition: a dotted line for a reference across component schemas, because D0 forbids one component from reading another's tables.

Figure type	Symbol	Meaning
Component map (Fig. 1)	Blue rounded box / gray rounded box / gray dashed box / ellipse	Component we build (Cn) / glue module (Gn) used by every backend component / external system (En) / actor
	Dotted outline	One deployable backend (D-01)
	Arrow labelled Ix; red label	Interface from caller to callee, reply on the same interface; red = revised in D2 (Part 2.10)
Data model (Fig. 2, 3)	Rounded box; dashed box	Entity in the schema of the component in its title; transient object never stored
	Solid line; dotted line	Foreign key within one schema; logical reference by ID across schemas (no database FK, kept current by events or I17a/I17b)
	1, 0..1, 0..*, 1..* at both ends; [PK] [FK] [UQ] [UQ*] ?	Cardinality read from the box next to the label; keys, partial unique index, nullable
State machine (Fig. 4, 5, 6)	Rounded box; filled dot; ringed dot	State stored in the named column; initial; final
	event [guard] / action; red arrow or note	Transition; error outcome that refuses the request and leaves the state unchanged
Activity with error paths (Fig. 7, 8)	Box; diamond; black arrow; red arrow and box; dashed arrow	Step (owner in brackets); decision; happy path; error path ending in its outcome (R recoverable, N needs a person); asynchronous hand-off through an outbox or queue row

Data conventions (unchanged from D1): dates ISO 8601 yyyy-mm-dd; datetimes UTC; business days and "today" in America/New_York; money in integer cents; all paths under `/api/v1`.

Error envelope and code rule (new in D2, R-08). Every error body is `{"error": {"code": "INVALID_INPUT", "message": "...", "field": "endDate"?, "requestId": "..."}}`, and 429/503 add a Retry-After header. 400 = the request fails schema or range validation (shared zod schemas). 401 = no valid session. 403 = valid session, not allowed. 404 = not found or not visible to the caller. 409 = conflicts with current state. 410 = expired. 422 = well-formed but refused by a business rule. 429 = rate limited. 503 = a dependency is down; retry.

1.2 Two-minute overview

The system. A web marketplace where UC students sublease apartments for a co-op term. Only people who prove they own an @mail.uc.edu mailbox can list, search, or message. Search returns only listings whose availability covers the whole co-op term. Every listing description is screened for Fair Housing preference language and held for a human reviewer if anything matches. When an owner picks a subleaser, the unit's property manager approves or rejects before move-in. D0 split this into eight components (C1 to C8) inside one deployable backend on PostgreSQL.

What D1 covered. C3 Listing Service (listing lifecycle, Fair Housing hold and review, transactional outbox) and C5 Search & Match (derived search index, GiST date-coverage query), with their data model, algorithms, API, and technology choices.

What D2 adds. (1) Data model, behavior, and interface for C1, C2, C4, C6, C7, C8 and the glue modules G1 Authentication, G2 Configuration, G3 Logging and audit (Part 2). (2) Four state machines (Account, Listing, ApprovalRequest, Notification plus outbox rows) and two error-path diagrams for the riskiest flows (Fig. 4 to 8). (3) Corrections to D1 found by checking it against D0, and revisions to D0 interfaces that leaked data or had loose types (Parts 2.8, 2.10). (4) 69 edge cases across all four categories for every component (Part 3), a forward and reverse traceability check (Part 4), ten rationale entries (Part 5), and the peer review notes (Part 6).

1.3 Component status

Component	Owner (D0)	Detailed in	Data model	Behavior	Interface
C1 Web Client	Mohamed Aidja	D2 Part 2.2	No persistent data	Session states, screen table	Consumes I2 to I6; error table
C2 Identity Service	Danvanth Vanganuri	D2 Part 2.3	Fig. 2	Algorithm C; Fig. 4, 8	Endpoints + I17a/I17b
C3 Listing Service	Jovan Jose Asker Fredy	D1; D2 Part 2.8 changes	D1 Fig. 1 + D2 changes	D1 Algorithm B; Fig. 4, 6,	D1 §5 + 2 GET endpoints

				7	
C4 Compliance Screener	Jovan Jose Asker Fredy	D2 Part 2.4	No persistent data (versioned file)	Algorithm D	screen() (I7)
C5 Search & Match	Jovan Jose Asker Fredy	D1; D2 Part 2.8	D1 Fig. 1	D1 Algorithm A	D1 §5
C6 Messaging Service	Mohamed Aidja	D2 Part 2.5	Fig. 3	Step lists	Endpoints
C7 Approval Service	Danvanth Vanganuri	D2 Part 2.6	Fig. 3	Fig. 5; Algorithm E	Endpoints
C8 Notification Service	Danvanth Vanganuri	D2 Part 2.7	Fig. 2	Fig. 6; Algorithm F	enqueue(); webhook
G1 Authentication middleware	Danvanth Vanganuri (proposed)	D2 Part 2.9	No persistent data	Step list	requireAuth(), verifyToken() (I18)
G2 Configuration	Jovan Jose Asker Fredy (proposed)	D2 Part 2.9	No persistent data (env + files)	Step list	getConfig() (I20)
G3 Logging and audit	Mohamed Aidja (proposed)	D2 Part 2.9	AuditLog (Fig. 2)	Step list	log(), audit.record() (I19), /healthz

1.4 Known gaps and open questions

ID	Gap or question	Current handling	Owner
GQ-01	Peer review feedback from the reviewing teams has not been merged into this draft.	Part 6.2 is the template; Part 5's "Changed by review" column to be updated after class.	All
GQ-02	Phrase list v1 content is not written. Ohio law (R.C. 4112) appears to add ancestry and military status to the federal protected classes; to confirm. Look-alike spellings outside the substitution map publish unflagged, and there is no report-a-listing feature.	Categories in Part 2.4 include both pending confirmation; residual risk accepted (R-04).	Jovan
GQ-03	A listing can never leave PUBLISHED: no withdraw, no filled-after-approval, no removal after its end date. No user story asks for it.	Not designed (reverse check, Part 4.2). Ask instructor whether to add a story.	Jovan
GQ-04	Managers and the unit registry are seeded by the team; a unit with no registered manager cannot complete US-03.	422 UNIT_NOT_REGISTERED with guidance (EC-C7-08). Self-service registration deferred (D-09).	Danvanth
GQ-05	Email provider (E1) not chosen: free-tier quota, webhook signing, and idempotency-key support unconfirmed.	Design needs all three (EC-C8-02, 04, 06); choose in Week 8.	Danvanth
GQ-06	Address normalization is rule-based, so unusual spellings can bypass the duplicate-listing 409.	Accepted (EC-C3-05); geocoding API rejected for cost.	Jovan
GQ-07	User_Stories did not number non-functional requirements; NFR-01 to NFR-07 are numbered here from D0 constraints. Priorities in Part 4 are the team's reading.	Align with User_Stories if it states other priorities.	All
GQ-08	A revoked verification or role change lasts until the 24 h token expires.	Accepted in D-03 (EC-G1-04).	Danvanth

2. Finished Components, the D2 Diagrams

Figure 1 shows every component D2 finishes, the glue modules, and the interfaces revised in Part 2.10.

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D2 Figure 1: component map with glue components and revised interfaces

Goal: Let UC students fill or find term-length subleases safely by limiting the marketplace to verified @mail.uc.edu students, matching listings to exact co-op dates, holding discriminatory listings for review, and giving property managers an approve/reject step before move-in.

Conventions: Blue rounded box = component we build (Cn). Gray rounded box = glue module (Gn) used by every backend component. Gray dashed box = external system (En). Ellipse = actor. Dotted outline = one deployable backend (D-01). Arrow Ix = interface from caller to callee (reply returns on it). Red label = interface revised in D2 (Part 2.10). Arrows from the G-band mean every backend component calls that glue module.

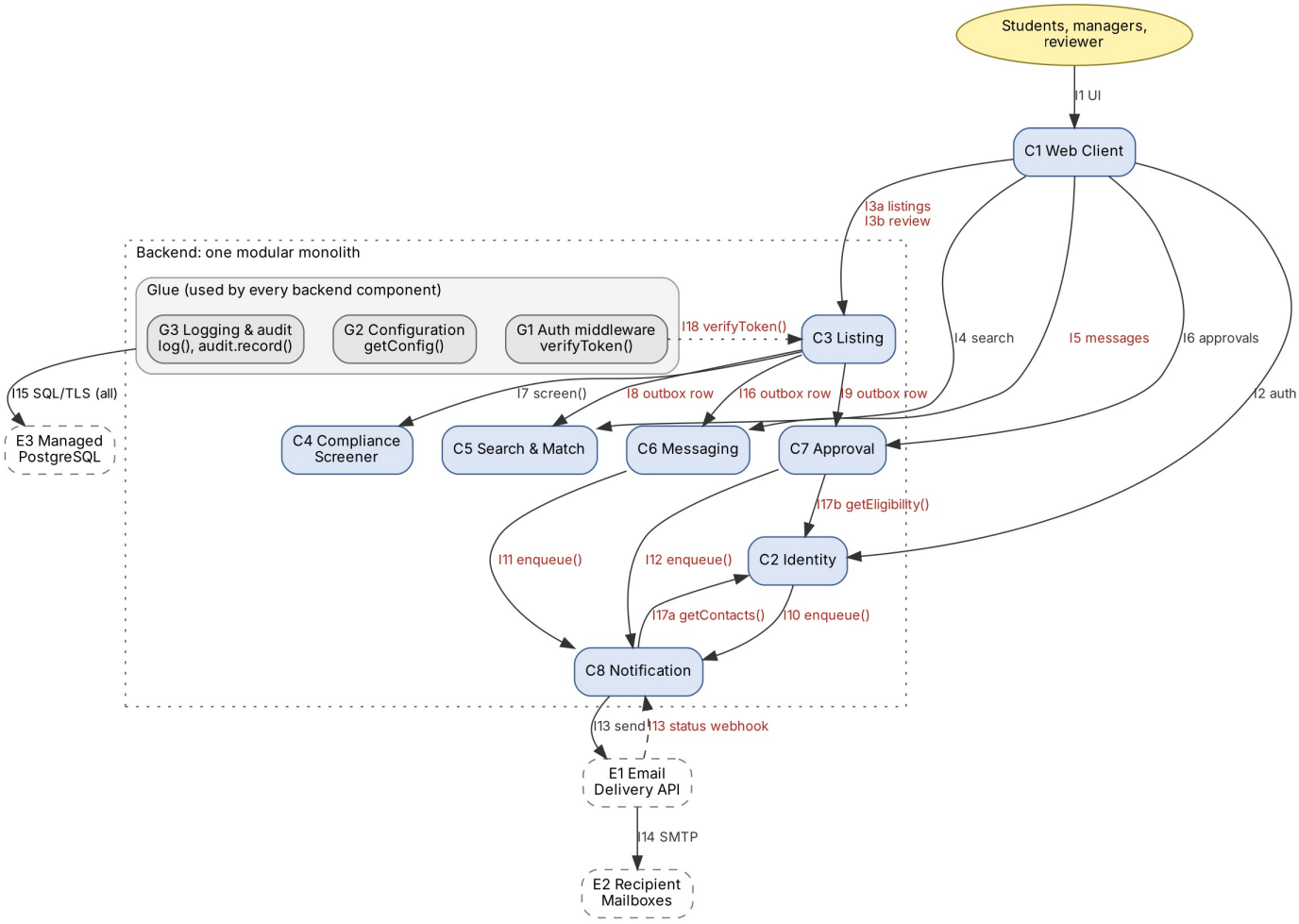


Figure 1. D2 component map. Glue modules G1 to G3 are called by every backend component; red labels are interfaces revised in D2.

2.1 Data models for the components finished in D2

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D2 Figure 2: data model for C2 Identity, C8 Notification, and G3 audit

Goal: Let UC students fill or find term-length subleases safely by limiting the marketplace to verified @mail.uc.edu students, matching listings to exact co-op dates, holding discriminatory listings for review, and giving property managers an approve/reject step before move-in.

Conventions: Rounded box = entity in the schema of the component named in its title. Dashed box = transient object, never stored. Solid line = foreign key inside one component's schema. Dotted line = logical reference by ID to another component's data (no cross-schema FK; kept current by events or the I17a/I17b lookups). Cardinality at both ends: 1, 0..1, 0..*, 1..*. [PK] primary key, [FK] foreign key, [UQ] unique, [UQ*] partial unique index, trailing ? = nullable. Datetimes UTC; money integer cents.

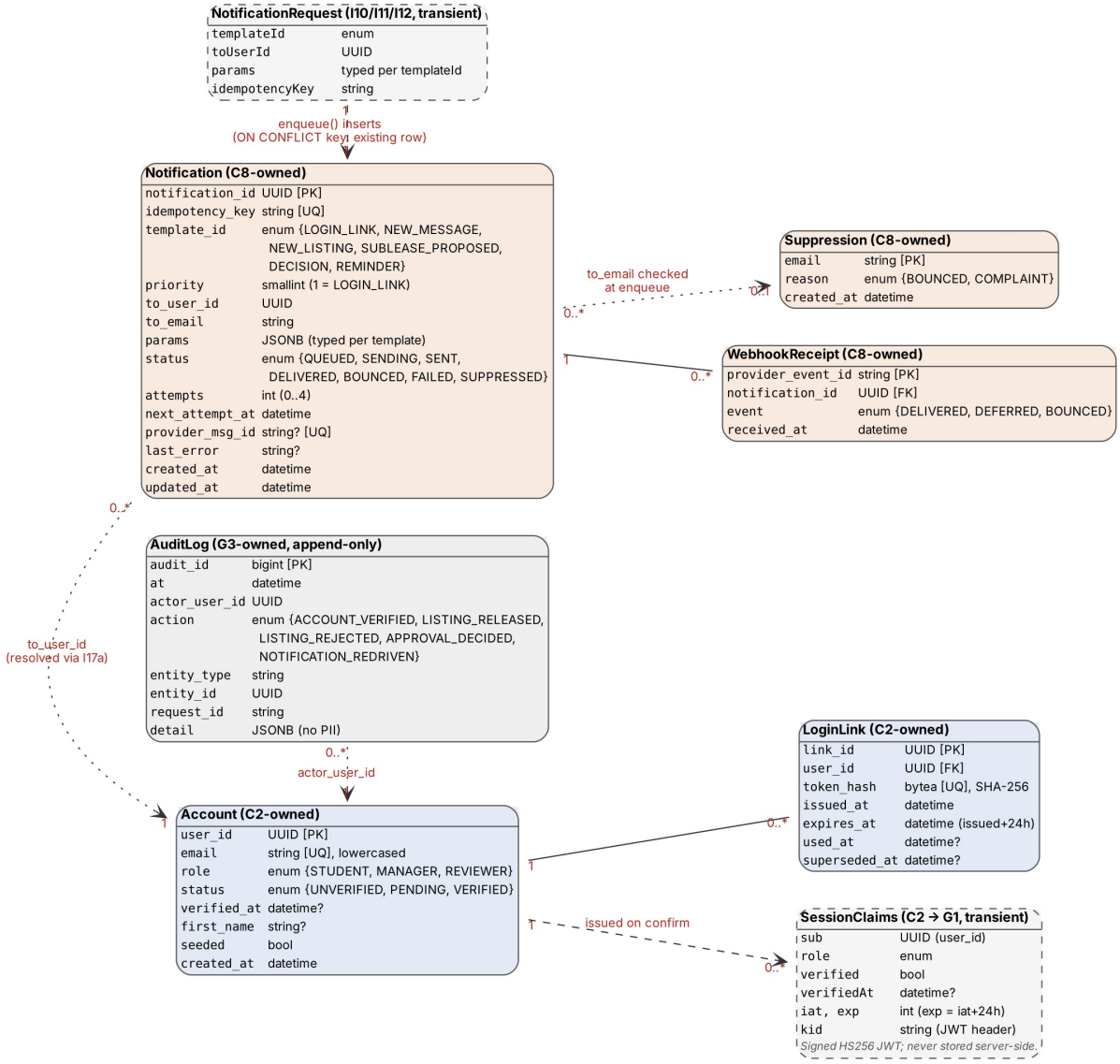


Figure 2. Data model for C2 Identity, C8 Notification, and G3 audit. Requirement trace: US-01, US-03, US-05; AC-01.2; NFR-02, NFR-03, NFR-05, NFR-06.

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D2 Figure 3: data model for C6 Messaging and C7 Approval

Goal: Let UC students fill or find term-length subleases safely by limiting the marketplace to verified @mail.uc.edu students, matching listings to exact co-op dates, holding discriminatory listings for review, and giving property managers an approve/reject step before move-in.

Conventions: Rounded box = entity in the schema of the component named in its title. Dashed box = transient object, never stored. Solid line = foreign key inside one component's schema. Dotted line = logical reference by ID to another component's data (no cross-schema FK; kept current by events or the I17a/I17b lookups). Cardinality at both ends: 1, 0..1, 0..*, 1..*. [PK] primary key, [FK] foreign key, [UQ] unique, [UQ*] partial unique index, trailing ? = nullable. Datetimes UTC; money integer cents.

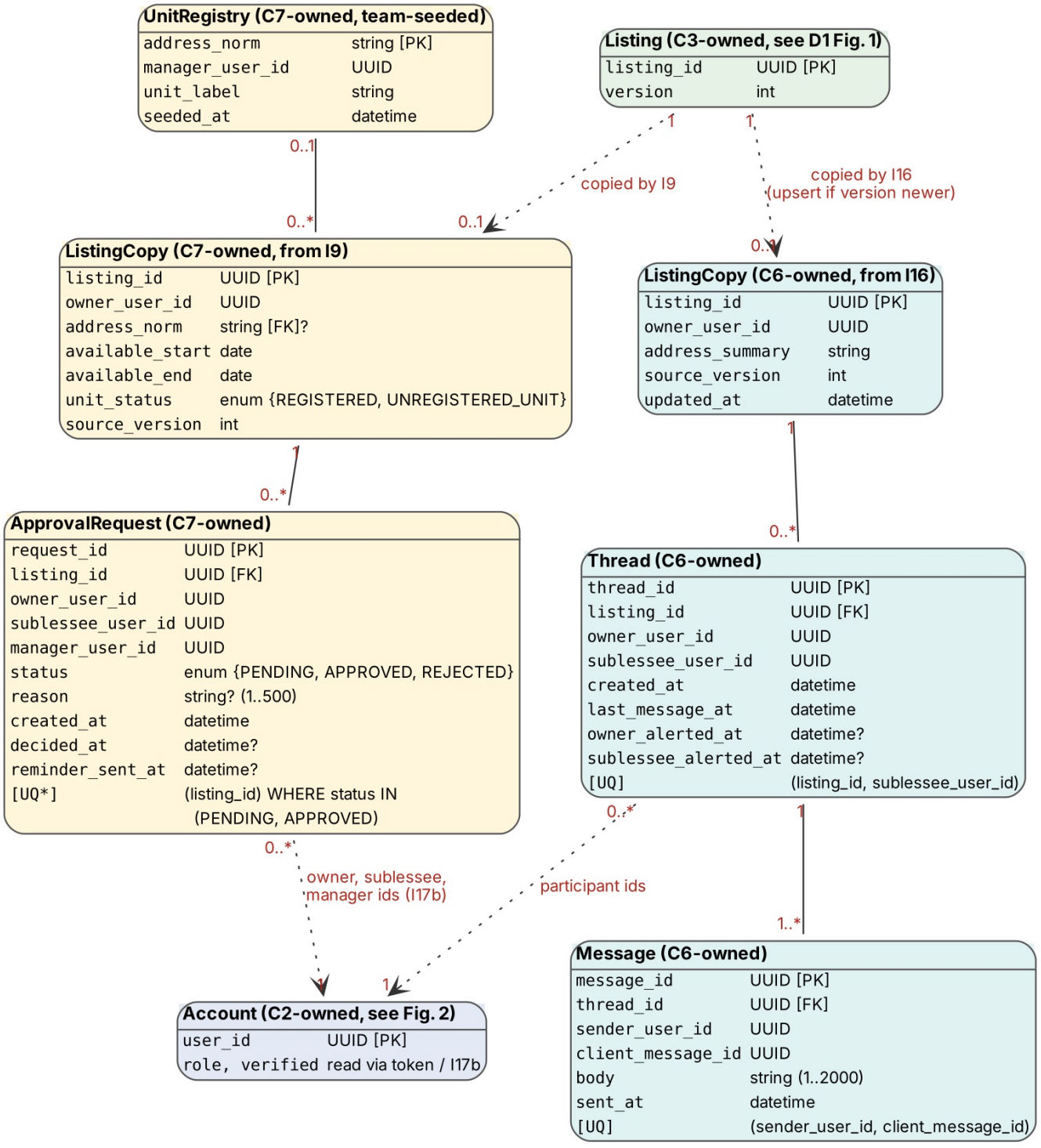


Figure 3. Data model for C6 Messaging and C7 Approval. Requirement trace: US-01, US-03, US-04; UC-02; AC-02.1, AC-02.2; NFR-05.

Decision	Choice and reason	Req.
Hash, not token	LoginLink stores SHA-256 of a 32-byte random token. A database read cannot be turned into a sign-in.	NFR-03
Status vs freshness	Account.status records whether the mailbox was ever proven; the 12-month rule is computed from verified_at at request time (G1), so no nightly job has to flip rows.	UC-01 A1
Copies, not cross-reads	C6 and C7 keep their own ListingCopy from I16 and I9, upserted only when source_version is newer (D1's guard), so neither reads C3's tables (D-11, D-12).	D-12
One thread per pair	Unique (listing_id, sublessee_user_id) makes the first message from a student create exactly one thread, even under concurrent sends.	US-01
Approval uniqueness	Partial unique index on listing_id WHERE status IN (PENDING, APPROVED): one live request per listing; after a rejection the owner may propose someone else.	US-03
Exactly one reminder	reminder_sent_at is both the record and the guard (Algorithm E).	AC-02.2
to_email on Notification	Resolved once at enqueue through I17a, so the worker needs no C2 call and the suppression check happens before the caller commits.	NFR-06
Append-only audit	AuditLog has INSERT only for the app role; detail never contains email or message text.	NFR-02, US-05

2.2 C1 Web Client

Data model: no persistent data. The browser keeps only the session token and its expiry in sessionStorage, and unsent form drafts in sessionStorage keyed by route (cleared on successful submit). All state lives in the backend.

Behavior: session states. Signed out → (link requested) Link sent → (Confirm pressed, 200) Signed in → (401 or token expiry) Signed out, draft kept → (sign in again) draft restored. 403 VERIFICATION_REQUIRED moves a signed-in student back to Link sent without losing the draft.

Screen (route)	Roles	Calls	Accessibility behavior
Sign in (/signin)	Anyone	POST /auth/link-requests	Single labelled email field; 429 shows a countdown announced politely
Confirm (/verify#token)	Anyone	POST /auth/link-confirmations	Token read from the URL fragment (never sent to logs); Confirm button is the first focus
New listing (/listings/new)	Verified student	POST /listings	Native date inputs with yyyy-mm-dd hint; character counter; flagged phrases listed in text, not colour
My listings (/listings/mine)	Verified student	GET /listings/mine	Status as text (Published, In review, Rejected)
Search (/search)	Verified student	GET /listings/search	Result count announced ("3 listings cover Jan 11 to Apr 30"); results are a list of headings
Listing (/listings/:id)	Verified student	GET /listings/{id}, POST /listings/{id}/messages	Message box labelled; send result announced
Messages (/threads, /threads/:id)	Verified student	GET /threads, GET and POST /threads/{id}/messages	New messages announced; "Propose as subleaser" button for owners
Approvals (/approvals)	Owner, sublessee, manager	GET, POST /approval-requests, POST /decision	Reason field required and labelled when Reject is chosen
Review queue (/review)	Reviewer	GET /review-queue, POST /listings/{id}/review	Flagged phrases listed beside the description

Interface. C1 offers I1 to users (HTML with ARIA landmarks, WCAG 2.1 AA) and consumes I2 to I6. It maps every error code to exactly one behavior:

Code	C1 behavior
400 INVALID_INPUT	Mark the field named in error.field, announce in the ARIA live region, move focus to the first invalid field
401 UNAUTHENTICATED	Save draft, route to /signin, restore draft and focus after sign-in (EC-C1-03)
403 VERIFICATION_REQUIRED / FORBIDDEN	Start re-verification / show "not allowed for your role" page
404	Not-found page with a link back to search
409	Inline message with the current state from the response and a Refresh action
410 EXPIRED	"This link has expired" with a Send new link button (AC-01.2)
422	Message next to the action naming the rule (for example UNIT_NOT_REGISTERED explains who to contact)
429, 503	Disable the action until Retry-After passes; GETs retry once after 2 s; POSTs never auto-retry (EC-C1-04)

Accessibility rules (US-04, NFR-01): every control is a native element with a visible label; every route change moves focus to the page heading; nothing is conveyed by colour alone; text contrast at least 4.5:1; all actions reachable by keyboard in visual order. Checked with axe-core in CI and a manual NVDA and VoiceOver pass on the search-and-message path.

2.3 C2 Identity Service

Data model: Account, LoginLink, SessionClaims (Fig. 2). **State machine:** Fig. 4. **Error paths:** Fig. 8.

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D2 Figure 4: state machines for Account (C2) and Listing (C3)

Goal: Let UC students fill or find term-length subleases safely by limiting the marketplace to verified @mail.uc.edu students, matching listings to exact co-op dates, holding discriminatory listings for review, and giving property managers an approve/reject step before move-in.

Conventions: Rounded box = state stored in the named column. Filled dot = initial state; ringed dot = final state. Arrow = transition, labelled event [guard] / action. Red arrow or note = error outcome: the request is refused with the named code and the state does not change. Every transition commits in one SQL transaction with its outbox or audit row.

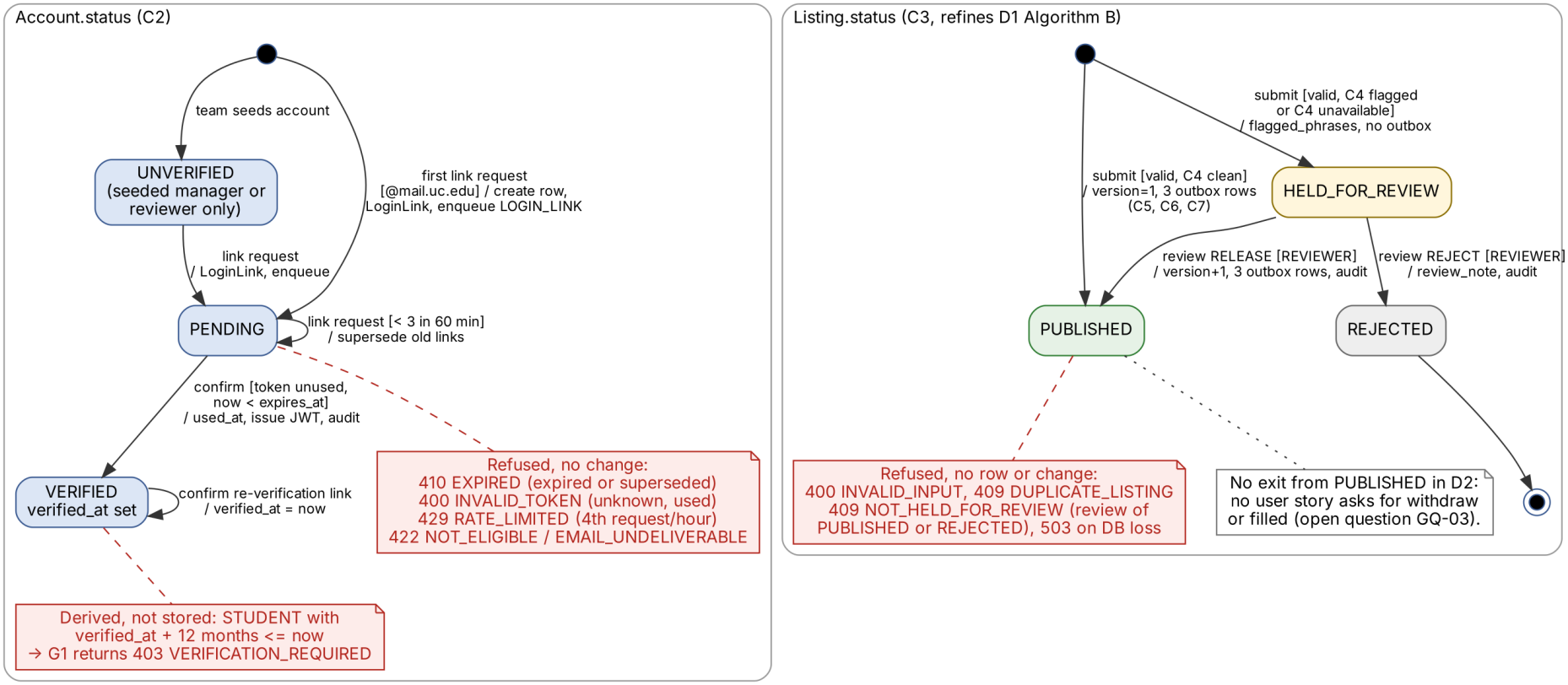


Figure 4. Account (C2) and Listing (C3) state machines, with error outcomes. Requirement trace: US-01, US-05; UC-01 A1; AC-01.2, AC-01.3, AC-01.4.

Algorithm C: one-time link request and confirmation

Part	Detail
What and why	Prove mailbox ownership for sign-up and every sign-in with one mechanism (D-02, D-09), and issue a 24 h session token carrying role and verification (D-03).
Inputs / outputs	Request in: email (3 to 254 chars). Out: 202 {status: PENDING}. Confirm in: token (43-char base64url), firstName? (1 to 50, required when the account has none). Out: 200 {status: VERIFIED, verifiedAt, role, sessionToken, expiresAt}.
Method (request)	1. Trim, lowercase, NFKC. 2. Eligible if the domain after the last @ is exactly mail.uc.edu with no + in the local part, or the address is a seeded account; else 422 NOT_ELIGIBLE. 3. If 3 LoginLink rows for this account were issued in the last 60 min: 429 with Retry-After. 4. BEGIN; upsert Account (new students start PENDING); set superseded_at on unused, unexpired links; insert LoginLink with SHA-256 of a fresh 32-byte random token and expires_at = now + 24 h. 5. enqueue(tx, LOGIN_LINK to the userId, link = APP_BASE_URL/verify#token). SUPPRESSED: ROLLBACK, 422 EMAIL_UNDELIVERABLE. Throws: ROLLBACK, 503. 6. COMMIT, 202.
Method (confirm)	1. Hash the token, look up LoginLink; none: 400 INVALID_TOKEN. 2. superseded or now ≥ expires_at: 410 EXPIRED. 3. firstName missing when needed: 400 INVALID_INPUT. 4. BEGIN; `UPDATE login_link SET used_at = now() WHERE link_id = $1 AND used_at IS NULL`; 0 rows: ROLLBACK, 400 INVALID_TOKEN. 5. Account status VERIFIED, verified_at = now (re-verification resets the 12-month clock), first_name if given; audit ACCOUNT_VERIFIED. COMMIT. 6. Sign HS256 JWT {sub, role, verified, verifiedAt, iat, exp = iat + 24 h} with the current kid.
Complexity	Constant work per call: one indexed count, a few row writes, one hash, one signature. Independent of account count.
Why this one	Token hash rather than token: a leaked table gives no sign-ins. POST to confirm rather than GET: mail scanners open links (R-03). Superseding old links: only the newest email works, so a stale forwarded email cannot sign in. Alternatives (UC SSO, passwords) were rejected in D-02 and D-09.
Edge cases	EC-C2-01 to EC-C2-08.

Endpoint or signature	Inputs	Success	Errors
POST /api/v1/auth/link-requests (no token)	email: string, required, 3 to 254	202 {status: "PENDING"}	400 INVALID_INPUT; 422 NOT_ELIGIBLE; 422 EMAIL_UNDELIVERABLE; 429 RATE_LIMITED + Retry-After; 503 NOTIFY_UNAVAILABLE
POST /api/v1/auth/link-confirmations (no token)	token: string, required, 43 chars; firstName: string 1 to 50, required if none stored	200 {status, verifiedAt, role, sessionToken, expiresAt}	400 INVALID_INPUT; 400 INVALID_TOKEN; 410 EXPIRED; 503 DB_UNAVAILABLE
getContacts(userIds: UUID[]) (I17a, C8 only)	1 to 100 ids	{userId, email, firstName}[]; unknown ids omitted	throws DbUnavailableError
getEligibility(userIds: UUID[]) (I17b, C7 only)	1 to 100 ids	{userId, role, verifiedFresh: bool}[]; unknown ids omitted	throws DbUnavailableError

2.4 C4 Compliance Screener

Data model: no persistent data. The phrase list is a versioned file in the repository, loaded once at startup by G2:

```
config/phrase_list.json
{ "version": "2026.10-v1",
  "entries": [ { "phrase": "no kids", "category": "FAMILIAL_STATUS" },
               { "phrase": "adults only", "category": "FAMILIAL_STATUS" }, ... ] }
category: RACE | COLOR | RELIGION | SEX | FAMILIAL_STATUS | NATIONAL_ORIGIN | DISABILITY
          | ANCESTRY | MILITARY_STATUS   (Ohio, pending confirmation, GQ-02)
          | OBFUSCATION                 (produced by the screener, never in the file)
```

Algorithm D: Fair Housing phrase screen

Part	Detail
What and why	Flag preference language before a listing can publish (US-05, NFR-04) and say exactly which phrase matched (AC-01.3).
Inputs / outputs	In: text, 0 to 2000 chars. Out: ScreenResult {flagged: bool, matches: [{phrase, category, start, end}], listVersion: string}. Throws ScreenerUnavailableError when no list is loaded.
Method	1. Build a normalized copy: NFKC, lowercase, curly quotes to straight, whitespace runs to one space, keeping an index map back to original offsets. 2. Build a folded copy with common substitutions (0→o, 1→i, 3→e, 4→a, 5→s, 7→t, @→a, $→s). 3. For each entry (normalized at load), find every occurrence in both copies where the characters before and after are not letters or digits (text start and end count as boundaries). 4. Any token mixing Latin with Cyrillic or Greek letters adds a match with category OBFUSCATION. 5. Map matches to original offsets, remove duplicates, sort by start. flagged = matches is not empty.
Complexity	Literal search: at most 2000 characters x about 300 entries, under 5 ms in Node. No regular expression is built from data, so there is no backtracking blow-up and the 2 s I7 budget cannot be reached by input.
Why this one	See R-04. A regex list risks ReDoS; a hosted moderation API sends listing text to a third party and adds cost; an ML classifier needs labelled data and cannot explain a miss (D-04).
Edge cases	EC-C4-01 to EC-C4-05; caller behavior EC-C3-06.

Interface (I7, revised): `screen(text: string): ScreenResult`, throws `ScreenerUnavailableError`. C3 wraps the call with a 2 s timer and treats a throw or timeout as SCREENER_UNAVAILABLE. D0's single `offset` becomes `start`/`end` so C1 can highlight the phrase.

2.5 C6 Messaging Service

Data model: ListingCopy, Thread, Message (Fig. 3).

Behavior: step lists

Start or continue a thread (POST /listings/{id}/messages): 1. G1 requires a verified, fresh caller. 2. Validate body (trim, 1 to 2000) and clientMessageId. 3. Load ListingCopy; missing: 403 LISTING_UNAVAILABLE. 4. Caller is the owner: 422 SELF_MESSAGE. 5. BEGIN; insert Thread ON CONFLICT (listing_id, sublessee_user_id) DO NOTHING, then select it; insert Message ON CONFLICT (sender_user_id, client_message_id) DO NOTHING (a retry returns the original); update last_message_at; COMMIT. 6. After commit, if the recipient's alerted_at is null or older than 15 min:

enqueue NEW_MESSAGE {listingId, threadId, senderFirstName} and set alerted_at; enqueue failure logs WARN (EC-C6-06). 7. 201.

Reply (POST /threads/{id}/messages): same steps, but step 3 loads the thread and returns 404 THREAD_NOT_FOUND if the caller is not a participant. **Consume I16:** upsert ListingCopy by listing_id when source_version is newer; otherwise ignore and ack.

Endpoint	Inputs	Success	Errors
POST /api/v1/listings/{id}/messages	id: UUID path; body: string 1 to 2000; clientMessageId: UUID	201 {threadId, messageId, sentAt}	400 INVALID_INPUT; 401; 403 VERIFICATION_REQUIRED; 403 LISTING_UNAVAILABLE + Retry-After; 422 SELF_MESSAGE; 503
POST /api/v1/threads/{id}/messages	id: UUID path; body; clientMessageId	201 {threadId, messageId, sentAt}	400; 401; 403 VERIFICATION_REQUIRED; 404 THREAD_NOT_FOUND; 503
GET /api/v1/threads	none	200 {threads: [{threadId, listingId, addressSummary, otherFirstName, role: OWNER|SUBLESSEE, lastMessageAt}]}	401; 403 VERIFICATION_REQUIRED; 503
GET /api/v1/threads/{id}/messages	before?: datetime; limit?: 1 to 50 (default 50)	200 {messages: [{messageId, senderFirstName, mine: bool, body, sentAt}]}	400; 401; 403; 404 THREAD_NOT_FOUND; 503

2.6 C7 Approval Service

Data model: UnitRegistry, ListingCopy, ApprovalRequest (Fig. 3). **State machine:** Fig. 5.

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D2 Figure 5: state machine for ApprovalRequest (C7, US-03, UC-02)

Goal: Let UC students fill or find term-length subleases safely by limiting the marketplace to verified @mail.uc.edu students, matching listings to exact co-op dates, holding discriminatory listings for review, and giving property managers an approve/reject step before move-in.

Conventions: Rounded box = state stored in the named column. Filled dot = initial state; ringed dot = final state. Arrow = transition, labelled event [guard] / action. Red arrow or note = error outcome: the request is refused with the named code and the state does not change. Every transition commits in one SQL transaction with its outbox or audit row.

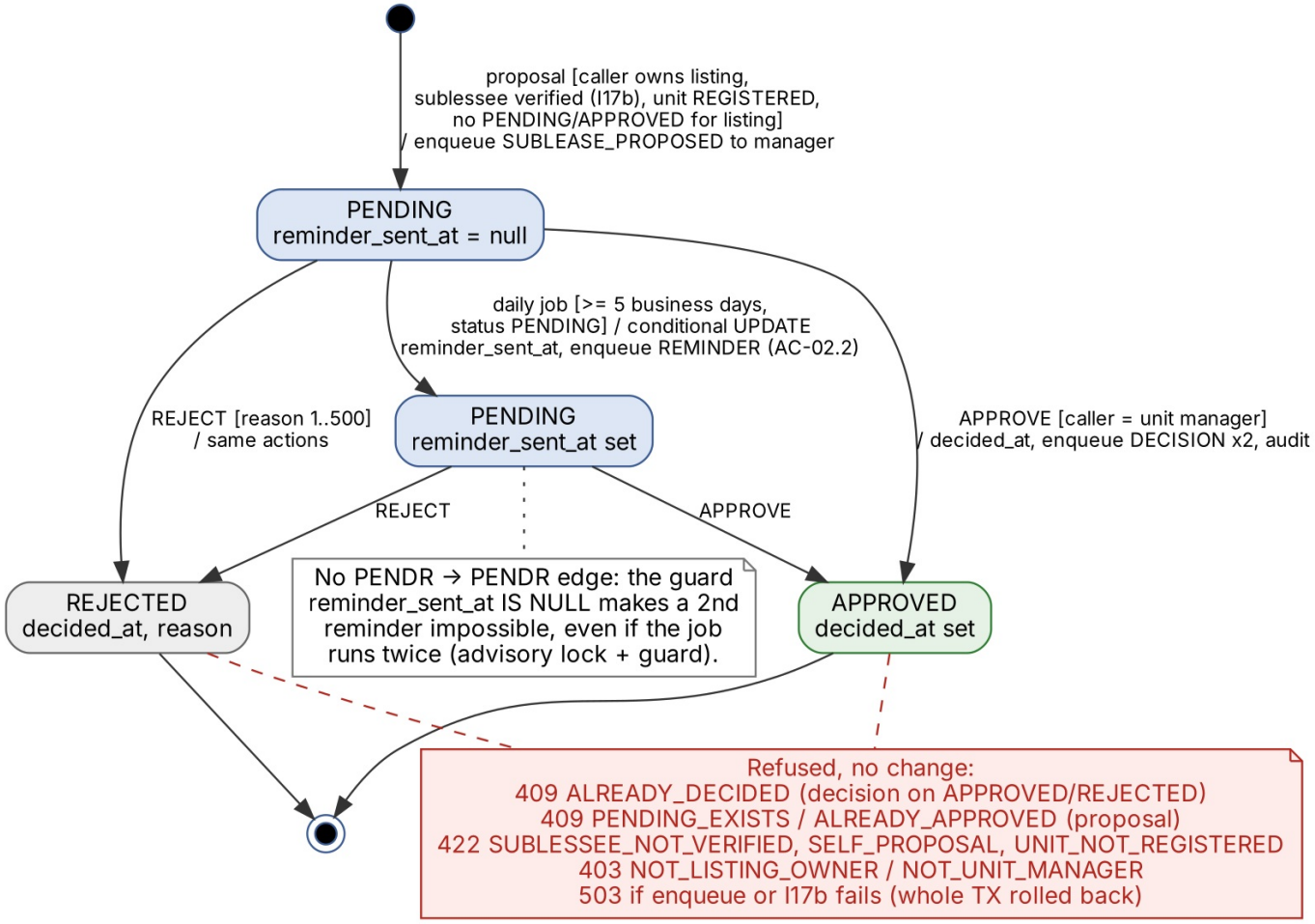


Figure 5. ApprovalRequest state machine. Requirement trace: US-03; UC-02; AC-02.1, AC-02.2.

Consume I9: upsert ListingCopy if source_version is newer; look up UnitRegistry by address_norm; set unit_status. If REGISTERED and this is the first copy for the listing, enqueue NEW_LISTING to the manager (key `NEW_LISTING:{listingId}`) in the same transaction; if UNREGISTERED_UNIT, ack without a notice (D0 I9).

Proposal: 1. G1 verified caller. 2. Load ListingCopy (missing: 409 LISTING_NOT_READY); caller must be owner_user_id (403 NOT_LISTING_OWNER). 3. sublessee ≠ owner (422 SELF_PROPOSAL). 4. unit_status REGISTERED (422 UNIT_NOT_REGISTERED). 5. getEligibility([sublesseeUserId]) must return verifiedFresh (422 SUBLESSEE_NOT_VERIFIED). 6. BEGIN; insert PENDING (partial unique index: 409 PENDING_EXISTS or ALREADY_APPROVED); enqueue SUBLEASE_PROPOSED to the manager; COMMIT; 201.

Decision: 1. G1 verified caller with role MANAGER. 2. BEGIN; `SELECT ... FOR UPDATE`; caller must be manager_user_id (403 NOT_UNIT_MANAGER). 3. `UPDATE ... SET status, reason, decided_at = now() WHERE request_id = $1 AND status = 'PENDING'`; 0 rows: 409 ALREADY_DECIDED. 4. enqueue DECISION to owner and sublessee (keys `DECISION:{requestId}:{userId}`); audit APPROVAL_DECIDED. 5. COMMIT; 200. Any failure in 3 or 4 rolls back (503).

Algorithm E: exactly-one reminder (AC-02.2)

Part	Detail
What and why	Send one reminder to the manager when a request has been PENDING for 5 business days, never two (AC-02.2).
Inputs / outputs	In: clock, holidays.json. Out: REMINDER notices; reminder_sent_at set on each reminded request.
Method	Runs daily at REMINDER_RUN_AT (09:00 America/New_York) inside the Node process. 1. `pg_try_advisory_lock(7001)`; not acquired: exit (another instance runs it). 2. Select PENDING requests with reminder_sent_at null. 3. For each where businessDaysBetween(created_at, now) ≥ 5 (Mon to Fri, holidays skipped, counting from the next business day): BEGIN; `UPDATE ... SET reminder_sent_at = now() WHERE request_id = $1 AND status = 'PENDING' AND reminder_sent_at IS NULL`; if 1 row, enqueue REMINDER (key `REMINDER:{requestId}`); COMMIT. 4. Unlock.
Complexity	O(p) for p pending requests, a few dozen at campus scale; the date arithmetic is O(days) per request with a 1-year cap.
Why this one	See R-07. Three independent guards make a second reminder impossible: the advisory lock, the conditional UPDATE, and the idempotency key.
Edge cases	EC-C7-01, 02, 04, 05.

Endpoint	Inputs	Success	Errors
POST /api/v1/approval-requests	listingId: UUID; sublesseeUserId: UUID (taken from a C6 thread)	201 {requestId, status: PENDING, createdAt}	400; 401; 403 VERIFICATION_REQUIRED; 403 NOT_LISTING_OWNER; 409 LISTING_NOT_READY, PENDING_EXISTS, ALREADY_APPROVED; 422 SELF_PROPOSAL, UNIT_NOT_REGISTERED, SUBLESSEE_NOT_VERIFIED; 503
POST /api/v1/approval-requests/{id}/decision	decision: APPROVE | REJECT; reason: string 1 to 500, required for REJECT	200 {requestId, status, decidedAt}	400 INVALID_INPUT; 401; 403 NOT_UNIT_MANAGER; 404; 409 ALREADY_DECIDED; 503
GET /api/v1/approval-requests	none (results limited to requests where the caller is owner, sublessee, or manager)	200 {requests: [{requestId, listingId, unitLabel, status, myRole, createdAt, decidedAt?}]}	401; 403 VERIFICATION_REQUIRED; 503

2.7 C8 Notification Service

Data model: Notification, Suppression, WebhookReceipt (Fig. 2). **State machine:** Fig. 6.

Verified Student Sublease Marketplace

D2 Figure 6: state machines for Notification (C8) and ListingEvent outbox rows (C3)

Goal: Let UC students fill or find term-length subleases safely by limiting the marketplace to verified @mail.uc.edu students, matching listings to exact co-op dates, holding discriminatory listings for review, and giving property managers an approve/reject step before move-in.

Conventions: Rounded box = state stored in the named column. Filled dot = initial state; ringed dot = final state. Arrow = transition, labelled event [guard] / action. Red arrow or note = error outcome: the request is refused with the named code and the state does not change. Every transition commits in one SQL transaction with its outbox or audit row.

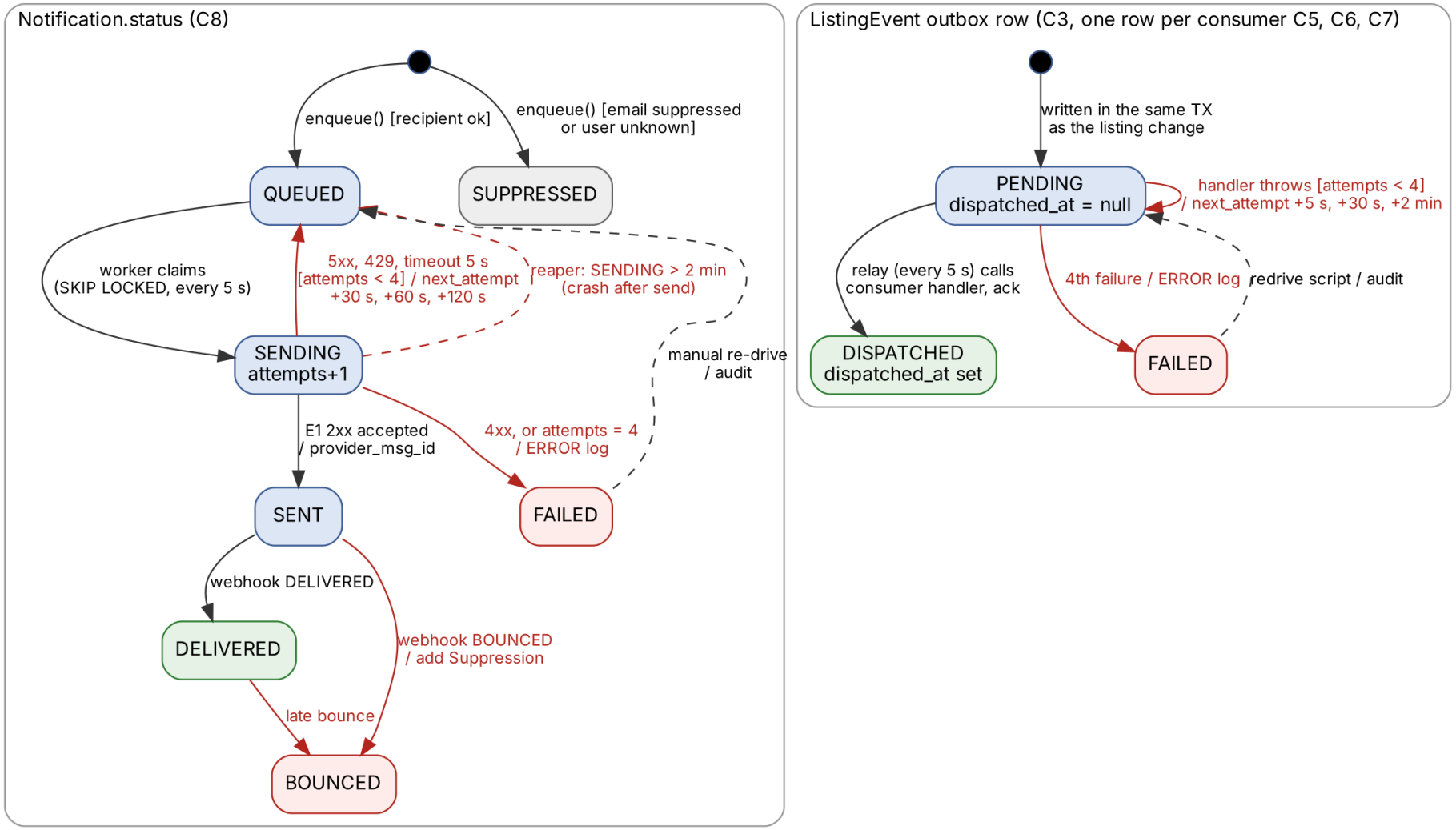


Figure 6. Notification and outbox-row state machines, including every retry and failure transition. Requirement trace: AC-01.1, AC-02.1; NFR-05, NFR-06.

Algorithm F: send worker

Part	Detail
What and why	Deliver every outbound email within the 5-minute budgets without sending duplicates or flooding UC Mail (AC-01.1, AC-02.1, NFR-05, NFR-06).
Method	Every NOTIFY_POLL_MS (5 s): 1. Reset rows SENDING for over 2 min to QUEUED (reaper). 2. Claim up to 20 rows `WHERE status = 'QUEUED' AND next_attempt_at <= now() ORDER BY priority, created_at FOR UPDATE SKIP LOCKED`; set SENDING, attempts + 1. 3. Render the template; POST to E1 with Idempotency-Key = idempotency_key and tag notificationId, 5 s timeout. 4. 2xx: SENT, provider_msg_id, clear params.link. 5xx, 429, timeout with attempts < 4: QUEUED, next_attempt_at = now + [30, 60, 120] s (or Retry-After if longer). 4xx or 4th attempt: FAILED, last_error, clear params.link, ERROR log.
Webhook	POST /api/v1/webhooks/email: 1. Verify HMAC signature and timestamp within 5 min, else 401. 2. Insert WebhookReceipt; duplicate provider_event_id: 200 no-op. 3. Find the row by the notificationId tag; unknown: 200. 4. Advance status only forward (SENT < DELIVERED < BOUNCED); DEFERRED changes nothing; BOUNCED inserts Suppression(email). 5. 200.
Complexity	One indexed claim query per cycle; at most 20 sends in flight. A few hundred emails a day is far below one worker's capacity.
Why this one	Same SKIP LOCKED pattern as D1's outbox, so one technique for both queues and no broker (D-05, NFR-07).
Edge cases	EC-C8-01 to EC-C8-09.

Interface	Inputs	Success	Errors
enqueue(tx, req) (I10, I11, I12, revised)	tx: caller's transaction; req: {templateId, toUserId, params (typed per template), idempotencyKey}	{notificationId, state: QUEUED | SUPPRESSED}; existing row if the key was seen	throws TemplateParamError; throws EnqueueError (database); caller decides (C2 503, C6 log, C7 503)
POST /api/v1/webhooks/email (I13 in)	headers: signature, timestamp; body: {eventId, messageId, tag, event: DELIVERED | DEFERRED | BOUNCED, timestamp}	200 (also for duplicates and unknown tags)	400 malformed; 401 bad signature or stale timestamp

templateId	params (typed)	Recipient	idempotencyKey	Priority
LOGIN_LINK	{link: URL, expiresAt: datetime}	the requesting account	LOGIN_LINK:{linkId}	1
DECISION	{requestId, decision, unitLabel}	owner and sublessee	DECISION:{requestId}:{userId}	2
SUBLEASE_PROPOSED	{requestId, unitLabel}	manager	PROPOSED:{requestId}	2
REMINDER	{requestId, unitLabel, businessDaysPending}	manager	REMINDER:{requestId}	2
NEW_LISTING	{listingId, unitLabel, availableStart, availableEnd}	manager	NEW_LISTING:{listingId}	3
NEW_MESSAGE	{listingId, threadId, senderFirstName}	other participant	NEW_MESSAGE:{threadId}:{userId}:{15-min window}	3

2.8 Changes to C3 and C5 since D1

D1 is not edited. These items supersede the named D1 parts; each traces to a self-audit item in Part 6.1 and a rationale entry.

#	Change	Supersedes	Why	Ref
1	ListingEvent gains consumer {C5_INDEX, C6_COPY, C7_APPROVAL}, status {PENDING, DISPATCHED, FAILED}, attempts, next_attempt_at, last_error. Each publish or release writes three rows; each is retried at +5 s, +30 s, +2 min, then FAILED (Fig. 6).	D1 Fig. 1 ListingEvent; Algorithm B outbox drain	D1's drain skipped I16 and I9; per-row isolation	R-01, PR-S02
2	Relay polls every 5 s (OUTBOX_POLL_MS), not 30 s.	D1 Algorithm B	30 s plus one retry misses the 60 s budget	R-02, PR-S01
3	Listing gains hold_reason {FLAGGED, SCREENER_UNAVAILABLE}? and screen_list_version.	D1 Fig. 1 Listing	Reviewer sees why a listing is held; verdict tied to a list version	EC-C4-02
4	New GET /api/v1/listings/{id} (PUBLISHED listings to verified callers; any status to the owner; else 404) and GET /api/v1/listings/mine.	D1 §5	No read endpoint existed for the listing page or owner status	PR-S11
5	endDate must be after today (America/New_York).	D1 §5 POST /listings	A listing that ended can never match	EC-C3-02
6	Review RELEASE and REJECT write an AuditLog row in the same transaction.	D1 Algorithm B review	Who decided is recorded	R-10
7	C5 unchanged; its I8 handler is the C5_INDEX consumer above.	none		

2.9 Glue components

G1 Authentication middleware

Data model: no persistent data (signing keys come from G2). **Behavior:** 1. Read `Authorization: Bearer <jwt>`; missing or malformed: 401. 2. `jwtVerify` (jose) with algorithms [HS256] and the key for the header kid (current or previous); any failure: 401 with no detail. 3. Build claims. 4. verifiedFresh = verified and (role ≠ STUDENT or verifiedAt + 12 months > now). 5. Route requires verified and not verifiedFresh: 403 VERIFICATION_REQUIRED; route requires a role the claims lack: 403 FORBIDDEN. 6. Attach claims to the request. Applied to every route except link-requests, link-confirmations, webhooks, and /healthz.

Interface (I18): `requireAuth(opts?: {verified?: boolean; roles?: Role[]}): RequestHandler`; `verifyToken(jwt: string): Claims` throws `AuthError{code: UNAUTHENTICATED}`. Edge cases EC-G1-01 to 05.

G2 Configuration

Data model: no persistent data in the database. Secrets come from host environment variables; everything else is a versioned file in `config/`, so a change is a reviewed commit and a deploy. **Behavior:** at process start, parse all sources with one zod schema; on any failure log FATAL with the key names only and exit 1, so the host keeps the previous deploy serving. The result is frozen; there is no runtime reload. Seeds are applied by a migration script, not at startup.

Key	Type and range	Used by

DATABASE_URL	postgres URL, required	all (I15)
JWT_KEYS, JWT_CURRENT_KID	JSON {kid: secret ≥ 32 bytes}, 1 or 2 keys; kid present in JWT_KEYS	C2, G1
EMAIL_API_KEY, EMAIL_WEBHOOK_SECRET, EMAIL_FROM	strings, required	C8
APP_BASE_URL	https URL	C2 (link), C8 (templates)
OUTBOX_POLL_MS, NOTIFY_POLL_MS	int 1000 to 10000, default 5000	C3 relay, C8 worker
SEND_TIMEOUT_MS / SCREEN_TIMEOUT_MS	int 1000 to 10000, default 5000 / int 500 to 2000, default 2000	C8 / C3
REMINDER_RUN_AT	HH:MM in America/New_York, default 09:00	C7
LOG_LEVEL	debug | info | warn | error	G3
config/phrase_list.json, holidays.json, seed_accounts.json, seed_units.json	schemas in Part 2.4; [date]; [{email, role, firstName}]; [{addressNorm, managerEmail, unitLabel}]	C4; C7; C2 and C7 migrations

Interface (I20): `getConfig(): Readonly<Config>`; `loadConfig(env, dir): Config` throws `ConfigError{keys: string[]}`. Edge cases EC-G2-01 to 04.

G3 Logging and audit

Data model: AuditLog (Fig. 2). Operational logs are not stored by us: JSON lines to stdout, kept by the host's log stream. **Behavior:** request middleware assigns requestId (also returned in every error body) and writes one line per request {at, level, requestId, method, route, status, ms, userId?}. Components log named events with allow-listed fields only; email becomes a short hash; tokens, links, message bodies, and descriptions are never logged. audit.record inserts in the caller's transaction. GET /healthz runs `SELECT 1` and returns 200 {db: "ok"} or 503.

Interface (I19): `log.info|warn|error(event: string, fields: Partial<Record<AllowedKey, string | number | boolean>>): void` (never throws); `audit.record(tx, {actorUserId, action, entityType, entityId, detail}): Promise<void>` throws on failure so the caller rolls back. Operations: `npm run redrive -- --outbox|--notification <id>` resets a FAILED row and writes NOTIFICATION_REDRIVEN. Edge cases EC-G3-01 to 04.

2.10 Revisions to D0 interfaces

Interface	Problem in D0	Revision in D2
I2	No field supplied the first name that I11 sends; link-requests returned the verification status of any address, which reveals whether an account exists.	link-confirmations takes firstName when none is stored; link-requests always returns 202 PENDING for eligible addresses.
I3	One interface mixed student create and reviewer decisions; invalid input was 422 while D1 used 400.	Split into I3a (create, read, mine) and I3b (review queue, review). Validation failures are 400 (R-08).
I5	Empty or oversized body was 422; no idempotency; no reply path for the owner.	400 for body; clientMessageId for safe retries; POST /threads/{id}/messages for replies.
I6	REJECT without reason was 422.	400 INVALID_INPUT (schema rule); 422 kept for business refusals.
I7	Single offset; unavailability was implicit.	start and end offsets; explicit ScreenerUnavailableError.
I8, I9, I16	One event, with fan-out by the relay; eventId-based upsert; I9 sent the raw address, which C7 cannot match against its registry.	One outbox row per consumer; upsert by listing_id with source_version (D1 guard); I9 carries addressNorm and addressSummary.
I10, I11, I12	Three signatures; I10 passed a raw email; I12 params were an untyped object.	One enqueue(tx, NotificationRequest) addressed by toUserId, params typed per template (R-06).
I13	Webhook matched rows by provider messageId, which may not be saved yet.	Rows matched by the notificationId tag; signature plus 5-minute timestamp window; event-id dedupe.
I17	One lookup returned email, role, verified, firstName to both C7 and C8, so C7 received emails it never uses.	I17a getContacts (C8 only) and I17b getEligibility (C7 only, no email) (R-05).
New I18, I19, I20	Glue behavior was implicit in D0.	verifyToken (G1), log and audit.record (G3), getConfig (G2).

2.11 Timing budgets rechecked

D0's worst cases left out the send timeout and the poll delay after each backoff. Recomputed with a 5 s poll, 5 s send timeout, and the retry schedules above:

Requirement	Budget	Worst-case path	D2 worst case
AC-01.1 search	60 s	relay poll 5 + C5 fails, wait 5 + poll 5 + fails, wait 30 + poll 5 + succeeds	50 s (D0 said 41 s)
AC-01.1 notice	5 min	relay 5 + C7 enqueue + worker poll 5 + three failed sends (5 s timeout each) with waits 30, 60, 120 and a poll after each + 4th send	about 251 s = 4.2 min
AC-01.1 notice, both chains retrying	5 min	as above plus C7 delivery failing twice (adds 45 s)	about 296 s = 4.9 min; any further retry misses and is logged FAILED
AC-02.1	5 min	enqueue in decision TX + worker poll 5 + the send chain	about 246 s = 4.1 min (D0 said 3.5 min)
AC-02.2	1 reminder after 5 business days	first daily run after day 5	same day or next business morning

3. Edge Cases and Error Handling

Every component and glue module has at least one row in each of the four categories. **Behavior** is one of retry, degrade, fall back, or surface an error (or n/a where the design makes the case impossible or idempotent). Figures 6, 7, and 8 draw the error paths; Fig. 7 is the riskiest flow (legal exposure plus two timing budgets).

ID	Comp.	Category	Edge case	Design decision	Behavior	Handler	Recov.
EC-C1-01	C1	Boundary values	Description of exactly 2000 vs 2001 characters, including emoji (UTF-16 length).	C1 and C3 import the same zod schema package, so both count `string.length` identically. A live counter shows remaining characters; 2001 is blocked in the form and the server 400 stays authoritative.	Surface error	C1 form validator; C3 zod	Yes
EC-C1-02	C1	Timing and concurrency	Double-click or slow network sends POST /listings twice.	Submit button disabled while a request is in flight. If a second request still arrives, the partial unique index returns 409 DUPLICATE_LISTING and C1 opens My listings instead of showing an error.	Surface error	C1 submit handler; C3 index	Yes
EC-C1-03	C1	Bad or missing input	Session token expires (401) while a long form is open.	Draft saved to sessionStorage on every change. On 401 C1 routes to sign-in, then restores the draft and focus after the student returns from the link.	Fall back	C1 API client	Yes
EC-C1-04	C1	External dependencies	Backend unreachable or returns 503.	GET requests retry once after 2 s; POSTs never auto-retry. Error and a Retry button are announced in the ARIA live region; draft is kept.	Retry, then surface	C1 API client	Yes
EC-C2-01	C2	Boundary values	Link opened at exactly issued_at + 24 h.	Valid only while `now < expires_at` (strict). At or after expiry: 410 EXPIRED, account stays PENDING, C1 offers a new link (AC-01.2).	Surface error	C2 confirm handler	Yes
EC-C2-02	C2	Boundary values	4th link request inside 60 minutes.	Count LoginLink rows with issued_at in the last 60 min; at 3, return 429 RATE_LIMITED with Retry-After = seconds until the oldest of the three ages out.	Surface error	C2 link-request handler	Yes, after wait
EC-C2-03	C2	Boundary values	verified_at exactly 12 months ago, including a leap-day verification (2028-02-29).	Freshness uses Postgres `verified_at + interval '12 months'` (Feb 29 maps to Feb 28); stale when `now >=` that value. G1 then returns 403 VERIFICATION_REQUIRED.	Surface error	G1 middleware	Yes, re-verify
EC-C2-04	C2	Timing and concurrency	UC Mail (Microsoft 365) link scanning opens the emailed URL before the student does, or the student double-clicks.	The emailed URL loads a C1 page and never consumes the token on GET. Only the Confirm button's POST does, through `UPDATE ... SET used_at = now() WHERE used_at IS NULL`; a second POST gets 0 rows and 400 INVALID_TOKEN, and C1 continues if it already holds a session.	Surface error	C2 confirm handler; C1 verify page	Yes
EC-C2-05	C2	Bad or missing input	Email variants: upper case, spaces, `x@mail.uc.edu.evil.com`, `x@uc.edu`, `a+tag@mail.uc.edu`.	Trim, lowercase, NFKC; domain = text after the last `@` must equal `mail.uc.edu` exactly. A `+` in the local part is rejected (one account per UC mailbox). Otherwise 422 NOT_ELIGIBLE unless the address is a seeded account.	Surface error	C2 link-request handler	Yes
EC-C2-06	C2	Bad or missing input	First verification without firstName (needed by NEW_MESSAGE alerts).	link-confirmations requires firstName (1 to 50 chars) when Account.first_name is null; otherwise 400 INVALID_INPUT naming the field, token not consumed.	Surface error	C2 confirm handler	Yes
EC-C2-07	C2	External dependencies	C8 enqueue throws (database error) during a link request.	Whole transaction rolls back: no LoginLink, account not PENDING; 503 NOTIFY_UNAVAILABLE with Retry-After 30 (D0 I10).	Surface error	C2 link-request handler	Yes
EC-C2-08	C2	External dependencies	Address previously bounced (on C8 Suppression list).	enqueue() returns SUPPRESSED; C2 rolls back and returns 422 EMAIL_UNDELIVERABLE so the student is not left waiting for an email that will never arrive.	Surface error	C2 link-request handler	No (needs a working mailbox)
EC-C3-01	C3	Boundary values	rentCents 0, 1, 500000, 500001; endDate equal to startDate.	Accept 1 to 500000 inclusive and endDate > startDate. Anything else: 400 INVALID_INPUT, no row written (AC-01.4).	Surface error	C3 zod schema	Yes
EC-C3-02	C3	Boundary values	endDate today or in the past.	endDate must be after today in America/New_York; else 400 INVALID_INPUT. Stops listings that can never match a future term.	Surface error	C3 zod schema	Yes
EC-C3-03	C3	Timing and concurrency	Two owners submit the same normalized address at the same moment.	Partial unique index on address_norm (PUBLISHED, HELD_FOR_REVIEW) lets exactly one commit; the other gets 409 DUPLICATE_LISTING.	Surface error	C3 create handler	Yes
EC-C3-04	C3	Timing and concurrency	Two reviewers act on the same held listing.	`UPDATE ... WHERE status = 'HELD_FOR_REVIEW' AND version = :v`; the loser gets 0 rows and 409 NOT_HELD_FOR_REVIEW. Review queue refreshes.	Surface error	C3 review handler	Yes
EC-C3-05	C3	Bad or missing input	Same unit written differently (`100 Sample Street #3B` vs `100 Sample St, Apt 3B`).	address_norm: lowercase, strip punctuation, USPS suffix and unit abbreviations (street to st, # and apartment to apt). Variants the rules miss can create a duplicate listing; accepted (GQ-06).	Degrade	C3 normalizer	Yes (reviewer or owner)
EC-C3-06	C3	External dependencies	C4 throws, list not loaded, or exceeds the 2 s budget.	Fail closed: HELD_FOR_REVIEW with hold_reason SCREENER_UNAVAILABLE, no outbox rows; 201 tells the owner the listing is in review (D0 I7).	Fall back	C3 create handler	Yes (reviewer releases)
EC-C3-07	C3	External dependencies	A consumer handler (C5, C6, or C7) fails on all 4 attempts.	That outbox row becomes FAILED with last_error and an ERROR log; other consumers' rows are unaffected. `npm run redrive` resets it to PENDING and writes an audit row. AC-01.1 is missed for that listing.	Retry, then surface	C3 outbox relay	Yes, by a person
EC-C3-08	C3	External dependencies	Database connection lost mid-transaction.	Transaction rolls back (no listing without its outbox rows); 503 DB_UNAVAILABLE with Retry-After 5; pool reconnects (D0 I15).	Surface error	C3 data-access layer	Yes

EC-C4-01	C4	Boundary values	Empty description; phrase at offset 0 or at the last character; 2000-character text.	Empty text returns flagged=false. Word-boundary check treats start and end of text as boundaries. Scan is bounded (2000 chars x at most 300 entries, under 5 ms).	n/a (defined result)	C4 screen()	Yes
EC-C4-02	C4	Timing and concurrency	Phrase list changes while listings are held or published.	List loads only at process start, so one request sees one version. Each verdict stores listVersion on the Listing; published listings are not rescreened (GQ-02).	Degrade	C4 loader; C3	Yes
EC-C4-03	C4	Bad or missing input	Substring false positives (`Christmas` vs a religious term) and look-alike spellings (`no k1ds`).	Match whole words only. A second pass runs on a copy with common substitutions folded (0 to o, 1 to i, 3 to e, @ to a, $ to s). Spellings outside the map are missed and publish (residual risk, GQ-02).	Degrade	C4 screen()	No automatic recovery
EC-C4-04	C4	Bad or missing input	Homoglyphs: a word mixing Latin and Cyrillic or Greek letters.	Any token with mixed scripts is a match with category OBFUSCATION, so the listing is held for a person to read.	Fall back (hold)	C4 screen()	Yes (reviewer)
EC-C4-05	C4	External dependencies	phrase_list.json missing or invalid at deploy.	G2 validation fails and the new process exits; the host keeps the previous version serving. If screen() is ever reached uninitialized it throws ScreenerUnavailableError and C3 holds (EC-C3-06).	Fall back	G2 loader; C3	Yes
EC-C5-01	C5	Boundary values	Availability equals the term exactly; or is one day short at either end.	Inclusive range `[)`: exact equality matches; one day short does not (D-06).	n/a (defined result)	C5 Algorithm A	Yes
EC-C5-02	C5	Boundary values	page past the last page (page 7 of 3); page 51.	Past last page: 200 with results [] and the true total. page > 50 or < 1: 400 INVALID_INPUT.	Surface error	C5 search handler	Yes
EC-C5-03	C5	Timing and concurrency	Duplicate or out-of-order I8 delivery.	Upsert on listing_id; ignore when source_version is not newer (D1).	n/a (idempotent)	C5 I8 handler	Yes
EC-C5-04	C5	Timing and concurrency	Search runs before a just-published listing is indexed.	Not shown yet; visible within the 50 s worst case (Part 2.11), inside AC-01.1's 60 s.	Degrade (eventual)	C3 relay	Yes
EC-C5-05	C5	Bad or missing input	`end=2027-02-30`, missing end, end before start.	zod date parse fails: 400 INVALID_INPUT naming the field.	Surface error	C5 search handler	Yes
EC-C5-06	C5	External dependencies	Database slow (statement_timeout 5 s) or down.	503 DB_UNAVAILABLE, Retry-After 5; C1 retries the GET once (EC-C1-04).	Retry, then surface	C5 data-access layer	Yes
EC-C6-01	C6	Boundary values	Body empty, whitespace only, 2000 chars, 2001 chars.	Trim, then 1 to 2000 chars; else 400 INVALID_INPUT (revises D0 I5's 422, Part 2.10).	Surface error	C6 zod schema	Yes
EC-C6-02	C6	Timing and concurrency	Message to a listing whose I16 copy has not arrived yet.	403 LISTING_UNAVAILABLE with Retry-After 30 (D0 I5). Copy normally arrives in about 10 s.	Surface error	C6 send handler	Yes
EC-C6-03	C6	Timing and concurrency	Client retries a send after a timeout (same message twice).	Unique (sender_user_id, client_message_id): the retry returns the original messageId with 201, no duplicate row or alert.	n/a (idempotent)	C6 send handler	Yes
EC-C6-04	C6	Timing and concurrency	Two first messages from the same sublessee to the same listing at once.	Thread insert uses `ON CONFLICT (listing_id, sublessee_user_id) DO NOTHING` then select, so both land in one thread.	n/a (idempotent)	C6 send handler	Yes
EC-C6-05	C6	Bad or missing input	Owner messages their own listing; body contains HTML or script.	Owner starting a thread on own listing: 422 SELF_MESSAGE. Bodies are stored raw and rendered as text by React (escaped); alert emails never include the body.	Surface error	C6 send handler; C1	Yes
EC-C6-06	C6	External dependencies	C8 enqueue fails, or a busy thread would send many alert emails.	Message is committed first; alert enqueue is best effort and failures log WARN (D0 I11). At most one NEW_MESSAGE alert per recipient per thread per 15 min (owner_alerted_at / sublessee_alerted_at).	Degrade	C6 alert step	Yes
EC-C6-07	C6	Bad or missing input	Caller is not a participant of the thread (guessed threadId).	404 THREAD_NOT_FOUND (not 403), so thread ids cannot be probed.	Surface error	C6 thread handlers	n/a
EC-C7-01	C7	Boundary values	Request created Friday 17:00 before a UC holiday week; 5 business days.	businessDaysBetween counts Mon to Fri, skips dates in holidays.json, and starts the count the next business day in America/New_York. Reminder fires on the first daily run after day 5 completes.	n/a (defined result)	C7 reminder job	Yes
EC-C7-02	C7	Timing and concurrency	Manager decides while the reminder job processes the same request.	Both paths lock the row (`SELECT ... FOR UPDATE`). Reminder update requires status PENDING; whichever commits first wins, so at most one reminder and exactly one decision.	n/a (serialized)	C7 decision handler; reminder job	Yes
EC-C7-03	C7	Timing and concurrency	Two decision clicks or a stale page submits a second decision.	`UPDATE ... WHERE status = 'PENDING'`; 0 rows returns 409 ALREADY_DECIDED with the current status (D0 I6).	Surface error	C7 decision handler	Yes
EC-C7-04	C7	Timing and concurrency	Reminder job runs twice (overlapping deploy, manual rerun).	`pg_try_advisory_lock` per run plus `reminder_sent_at IS NULL` guard in the UPDATE: exactly one reminder (AC-02.2).	n/a (idempotent)	C7 reminder job	Yes
EC-C7-05	C7	Timing and concurrency	Daily job missed because the service was down.	Next run selects every overdue PENDING request, so the reminder is late but still sent once.	Degrade	C7 reminder job	Yes
EC-C7-06	C7	Bad or missing input	REJECT with no reason or > 500 chars; sublessee equals owner; sublessee unknown or unverified.	Missing or long reason: 400 INVALID_INPUT. Self-proposal: 422 SELF_PROPOSAL. Unknown or unverified sublessee (I17b omits or verified=false): 422 SUBLESSEE_NOT_VERIFIED.	Surface error	C7 handlers	Yes
EC-C7-07	C7	Bad or missing input	Proposal while another request is PENDING or APPROVED for	Partial unique index gives 409 PENDING_EXISTS or ALREADY_APPROVED. No ListingCopy yet: 409	Surface error	C7 proposal handler	Yes

			the listing; I9 copy not yet arrived.	LISTING_NOT_READY with Retry-After 30.			
EC-C7-08	C7	External dependencies	Unit has no registered manager.	I9 consumer records UNREGISTERED_UNIT, acks, sends no notice; listing stays published (D0 I9). Proposal returns 422 UNIT_NOT_REGISTERED; C1 tells the owner to ask the team to register the manager (GQ-04).	Degrade	C7 I9 handler; proposal handler	Yes, by a person
EC-C7-09	C7	External dependencies	C8 enqueue or I17b lookup fails during a proposal or decision.	Whole transaction rolls back; 503 with Retry-After 30; nothing recorded without its notice (D0 I12).	Surface error	C7 handlers	Yes
EC-C8-01	C8	Boundary values	4th send attempt fails.	Attempts at 0, +30 s, +60 s, +120 s (D0 I13). After the 4th: FAILED, last_error, ERROR log.	Retry, then surface	C8 worker	Yes, by redrive
EC-C8-02	C8	Boundary values	Provider quota reached (429 with Retry-After).	Treated like 5xx but waits max(Retry-After, schedule). If that pushes past the 4th attempt the row is FAILED and logged. Rows are claimed by priority, so LOGIN_LINK goes first.	Retry, then degrade	C8 worker	Yes
EC-C8-03	C8	Timing and concurrency	Same notice enqueued twice (outbox redelivery, handler retry).	idempotency_key is unique; enqueue returns the existing row and no second email is sent.	n/a (idempotent)	C8 enqueue()	Yes
EC-C8-04	C8	Timing and concurrency	Worker crashes after E1 accepted the email but before SENT is saved.	Reaper returns rows stuck in SENDING over 2 min to QUEUED. The resend carries the same idempotency key to E1; if the chosen provider ignores it, one duplicate email is possible (accepted, GQ-05).	Retry	C8 reaper	Yes
EC-C8-05	C8	Timing and concurrency	Webhook arrives before SENT is saved, twice, or out of order (DELIVERED after BOUNCED).	Webhook resolves the row by the notificationId tag sent with the email, not by provider id. Duplicate provider_event_id: 200 no-op. Status only moves forward by rank (SENT < DELIVERED < BOUNCED).	n/a (idempotent)	C8 webhook handler	Yes
EC-C8-06	C8	Bad or missing input	Webhook with a bad signature or a timestamp older than 5 min.	401, body discarded, WARN log (D0 I13). The 5 min window blocks replays.	Surface error	C8 webhook handler	n/a
EC-C8-07	C8	Bad or missing input	Template params missing or wrong type; recipient user unknown.	Params are typed per template and checked with zod in enqueue(): TemplateParamError to the caller. Unknown userId (I17a omits it): state SUPPRESSED, no send.	Surface error	C8 enqueue()	Yes (caller fixes)
EC-C8-08	C8	External dependencies	E1 rejects the API key (401) or is down for hours.	4xx: FAILED immediately, ERROR log. Down: retries then FAILED. Listings, messages, and approvals keep working; emails are delayed until a person runs redrive.	Degrade	C8 worker	Yes, by a person
EC-C8-09	C8	External dependencies	Database holds login links in plain text inside params.	After SENT or FAILED the worker clears params.link in the same UPDATE, so a database read cannot replay a live link.	n/a (prevention)	C8 worker	n/a
EC-G1-01	G1	Boundary values	Token exp equals now.	Expired when `now >= exp` (single server clock, no leeway): 401 UNAUTHENTICATED.	Surface error	G1	Yes
EC-G1-02	G1	Timing and concurrency	Signing key rotated while tokens are live.	JWT_KEYS holds current and previous kid; tokens signed with the previous key verify until they expire (24 h), then the old key is removed.	Fall back	G1; G2	Yes
EC-G1-03	G1	Bad or missing input	Missing header, `Bearer` without token, alg `none`, tampered payload, unknown kid.	jose verifies with algorithms allow-list [HS256] and known kids only: 401 UNAUTHENTICATED with no detail about which check failed.	Surface error	G1	Yes (sign in)
EC-G1-04	G1	Timing and concurrency	Verification revoked or role changed while a token is live.	Token remains valid until exp (at most 24 h). Accepted trade-off of D-03.	Degrade	G1	Yes, after 24 h
EC-G1-05	G1	External dependencies	No signing key configured at deploy.	G2 refuses to start the process; previous deploy keeps serving.	Fall back	G2	Yes
EC-G2-01	G2	Boundary values	OUTBOX_POLL_MS set to 0 or 60000.	Allowed 1000 to 10000 ms; a value outside the range fails validation at startup, because 30 s or more would break AC-01.1.	Surface error (startup)	G2 loader	Yes
EC-G2-02	G2	Timing and concurrency	Old and new instances overlap during a deploy.	Pollers use `FOR UPDATE SKIP LOCKED`; the reminder job takes an advisory lock, so no row is processed twice.	n/a (serialized)	C3, C7, C8 jobs	Yes
EC-G2-03	G2	Bad or missing input	Missing secret, malformed holidays.json, or seed file with an unknown role.	zod validation fails: FATAL log naming the key only (never the value), exit code 1.	Surface error (startup)	G2 loader	Yes
EC-G2-04	G2	External dependencies	Host secret store unavailable during deploy.	Deploy fails before traffic switches; the previous instance keeps serving.	Fall back	Host (Render)	Yes
EC-G3-01	G3	Boundary values	A log field is very large (2000-char message body, stack trace).	Message bodies and descriptions are never logged; any field over 1 KB is truncated with a marker.	Degrade	G3 logger	n/a
EC-G3-02	G3	Timing and concurrency	Audit insert fails inside a review or approval decision.	audit.record() runs in the caller's transaction; a failure rolls back the decision and returns 503, so no decision exists without its audit row.	Surface error	G3 audit.record()	Yes
EC-G3-03	G3	Bad or missing input	Code tries to log an email, token, or link.	Logger accepts only allow-listed keys; email is replaced by a short hash and tokens or links are dropped (NFR-02).	n/a (prevention)	G3 logger	n/a
EC-G3-04	G3	External dependencies	Host log stream unavailable or slow.	Logging writes to stdout without awaiting; lines may be lost. Audit rows live in the database and are unaffected.	Degrade	G3 logger	Yes

Verified Student Sublease Marketplace

D2 Figure 7: error paths in the riskiest flow, publish a listing and fan out its events (US-01, US-05, AC-01.1, AC-01.3, AC-01.4)

Goal: Let UC students fill or find term-length subleases safely by limiting the marketplace to verified @mail.uc.edu students, matching listings to exact co-op dates, holding discriminatory listings for review, and giving property managers an approve/reject step before move-in.

Conventions: Rounded box = processing step (owning component in brackets). Diamond = decision. Black arrows = happy path. Red arrows and red boxes = error paths, each ending in the stated outcome; R = recoverable (user or system can retry), N = not recoverable without a person. Dashed arrow = asynchronous hand-off through an outbox or queue row. Codes are HTTP status + error code.

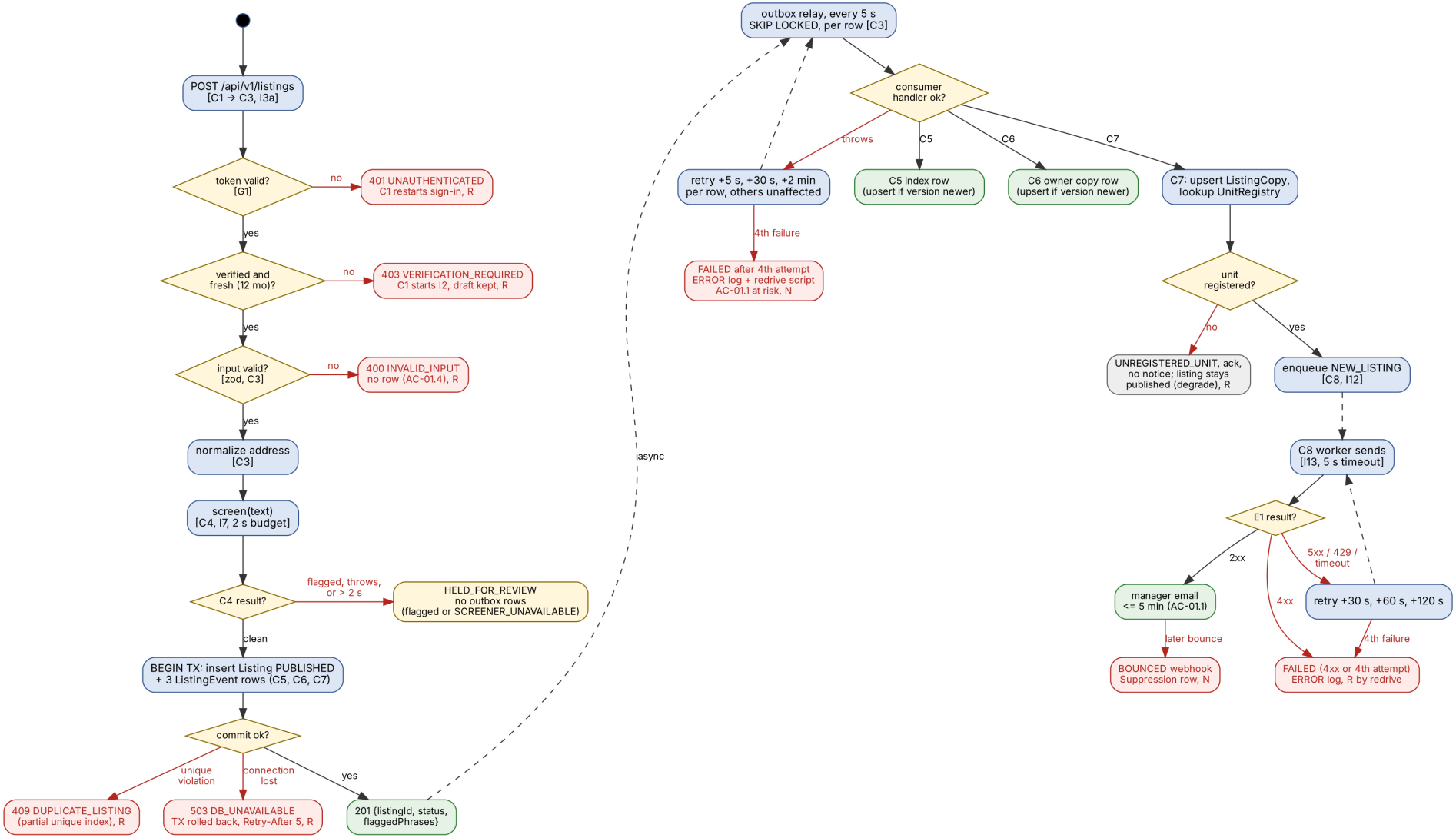


Figure 7. Publish flow with every decision and failure outcome. Requirement trace: US-01, US-05; AC-01.1, AC-01.3, AC-01.4; NFR-04, NFR-06. Edge cases EC-C3-01 to 08, EC-C4-05, EC-C7-08, EC-C8-01 to 08.

Verified Student Sublease Marketplace

D2 Figure 8: error paths in sign-in and verification (US-01, UC-01 steps 2 to 6, AC-01.2)

Goal: Let UC students fill or find term-length subleases safely by limiting the marketplace to verified @mail.uc.edu students, matching listings to exact co-op dates, holding discriminatory listings for review, and giving property managers an approve/reject step before move-in.

Conventions: Rounded box = processing step (owning component in brackets). Diamond = decision. Black arrows = happy path. Red arrows and red boxes = error paths, each ending in the stated outcome; R = recoverable (user or system can retry), N = not recoverable without a person. Dashed arrow = asynchronous hand-off through an outbox or queue row. Codes are HTTP status + error code.

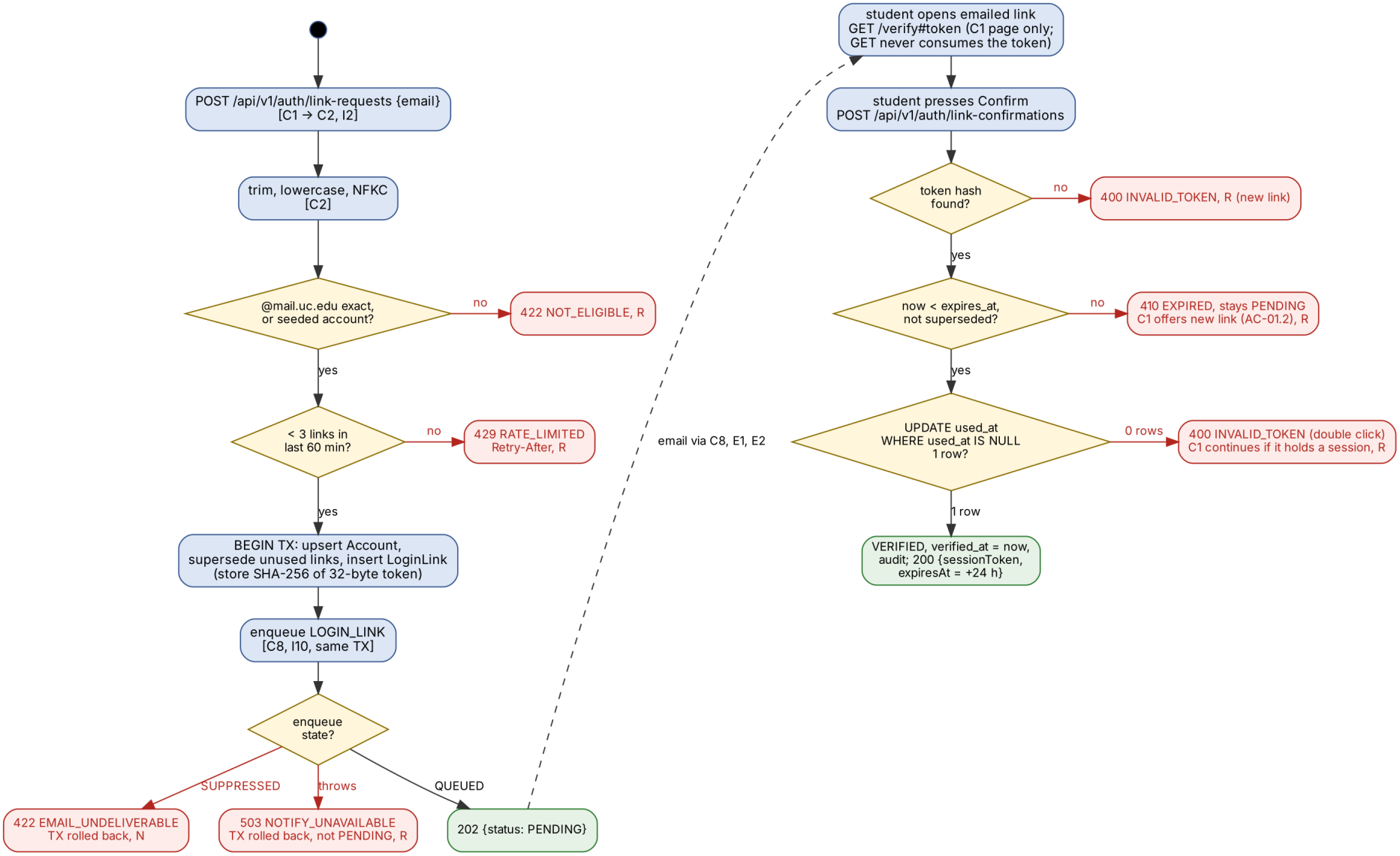


Figure 8. Sign-in and verification flow with error paths. Requirement trace: US-01; UC-01 steps 2 to 6; AC-01.2; NFR-03, NFR-05. Edge cases EC-C2-01 to 08.

4. Requirements Traceability

4.1 Forward check: requirement to design element

Story and criterion text is paraphrased from User_Stories and D0. Every row names an entity, endpoint, algorithm, or figure and cites edge-case IDs; the last column explains each gap.

ID	Requirement (short)	Priority	Design elements	How	Gap
US-01	Subleasing student lists a unit to verified UC students only; only verified students can contact them.	Must	Account, LoginLink (Fig. 2); POST /auth/link-requests and /link-confirmations; Algorithm C; G1 requireAuth; Listing + POST /listings (D1 §5); C6 POST /listings/{id}/messages; Fig. 4, 7, 8	Only an exact @mail.uc.edu mailbox can reach VERIFIED; every listing, search, and message route sits behind G1's verified-and-fresh check. EC-C2-01 to 08, EC-G1-01 to 05, EC-C6-02, 07.	None.
US-02	Sublessee finds listings that cover their exact co-op dates.	Must	SearchIndex.available daterange + GiST (D1 Fig. 1); D1 Algorithm A; GET /listings/search; GET /listings/{id} (Part 2.8)	Containment test `available @> [start, end)` returns only full coverage. EC-C5-01 to 06.	None.
US-03	Property manager approves or rejects a proposed subleaser before move-in.	Must	ApprovalRequest, UnitRegistry, ListingCopy (Fig. 3); Fig. 5; POST /approval-requests, /decision; GET /approval-requests	One request per listing in exactly one state; decision only by the registered manager. EC-C7-03, 06 to 09.	Managers and units are seeded by the team (D-09, D-11); self-service manager registration not designed (GQ-04).
US-04	Student using a screen reader or keyboard can search and message.	Must	C1 accessibility rules and error-presentation table (Part 2.2); I1 ARIA live region	Native form controls, landmarks, focus management, result counts announced; errors announced and focused. EC-C1-01, 03, 04.	Design states the rules; proof needs axe-core in CI plus a manual NVDA/VoiceOver pass, scheduled as a Week 8 task.
US-05	Listings with discriminatory preference language are held for Fair Housing review.	Must	C4 Algorithm D, PhraseList file, screen() (I7); Listing HELD_FOR_REVIEW (Fig. 4); review endpoints (D1 §5); AuditLog	Every description is screened before any row is published; flagged or screener failure means held, never published. EC-C3-04, 06, EC-C4-01 to 05.	Phrase list v1 content not written yet; look-alike spellings outside the map can pass (GQ-02).
UC-01	Verify account (steps 2 to 6) and publish a listing (7 to 10); A1 re-verify after 12 months.	Must	Fig. 8 (steps 2 to 6); Fig. 7 (steps 7 to 10); Account state machine (Fig. 4) VERIFIED self-loop; G1 freshness rule	Each use-case step maps to one box in Fig. 7 or 8; A1 is the derived stale condition in G1. EC-C2-03.	None.
UC-02	Owner proposes a sublessee; manager approves or rejects.	Must	Fig. 5; C7 proposal and decision step lists (Part 2.6)	Proposal takes sublesseeUserId from a C6 thread, so the owner never sees the sublessee's email. EC-C7-01 to 09.	None.
AC-01.1	Valid listing visible in verified search within 60 s and manager emailed within 5 min.	Must	ListingEvent per-consumer rows, relay every 5 s (Part 2.8, Fig. 6); C8 worker (Fig. 6); timing table (Part 2.11); Fig. 7	Worst case 50 s to search, 4.2 min to email (4.9 min if C7 delivery also retries). EC-C3-07, EC-C5-04, EC-C8-01, 02.	A 4th outbox or send attempt misses the budget; it is logged FAILED, not hidden.
AC-01.2	Link older than 24 h does not verify; no listing is published.	Must	LoginLink.expires_at; Algorithm C step 6; 410 EXPIRED; Fig. 8	Strict `now < expires_at`; account stays PENDING; G1 blocks listing creation. EC-C2-01.	None.
AC-01.3	Flagged listing is held, shows the flagged phrase, and returns 0 results in others' searches.	Must	Listing.flagged_phrases, HELD_FOR_REVIEW writes no outbox rows (Fig. 4, 7); SearchIndex holds only PUBLISHED (D1)	No outbox row means no index row; 201 returns flaggedPhrases. EC-C3-06, EC-C4-03, 04.	None. (D2 moves AC-01.3 citations for C4-unavailable holds here from D1.)
AC-01.4	Invalid dates or rent are rejected before any record is created.	Must	Shared zod schema; POST /listings 400 INVALID_INPUT (Fig. 7)	Validation runs before the transaction opens. EC-C3-01, 02, EC-C1-01.	D1 cited AC-01.4 for the 409 on reviewing a non-held listing; D2 follows D0's mapping (PR-S04).
AC-02.1	Approval sets APPROVED with a timestamp and both students are emailed within 5 min.	Must	ApprovalRequest.decided_at; DECISION enqueued twice in the decision transaction (Fig. 5); timing table	Enqueue in the same transaction means no decision without notices; worst case 4.1 min. EC-C7-09, EC-C8-01.	None.
AC-02.2	No manager response in 5 business days sends exactly 1 reminder; subleaser sees Pending.	Must	Algorithm E; reminder_sent_at; advisory lock; holidays.json (G2); GET /approval-requests	Conditional UPDATE plus lock gives exactly one reminder; GET shows PENDING to the sublessee. EC-C7-01, 02, 04, 05.	None.
NFR-01	Every screen meets WCAG 2.1 AA (D0 I1, Part 6).	Must	C1 accessibility rules (Part 2.2)	Same rules apply to every route in the C1 screen table.	Verification method as for US-04.
NFR-02	UC email addresses are never shown to other users (D0 Part 6, security).	Must	I17a (C8 only) and I17b (no email) split; thread responses expose senderFirstName only; G3 redaction	Only C8 ever reads an email address. EC-G3-03.	None.
NFR-03	Passwordless sign-in with short-lived signed sessions (D-02, D-03, D-09).	Must	SessionClaims, G1 verifyToken (I18), JWT_KEYS rotation, LoginLink.token_hash	Only token hashes are stored; sessions expire in 24 h. EC-C2-04, EC-G1-01 to 05.	Revocation waits up to 24 h (D-03, GQ-08).
NFR-04	No listing is published unscreened (Fair Housing, D0 Part 6).	Must	C3 fail-closed branch (Fig. 7); ScreenerUnavailableError; G2 refuses to start without the list	Every failure mode of C4 ends in HELD_FOR_REVIEW. EC-C3-06, EC-C4-05.	None.
NFR-05	Outbound email is rate-limited to protect UC Mail (D0 I2, E2).	Must	3 links/hour (Algorithm C); alert throttle columns on Thread (Fig. 3); C8 priority claim	Caps the two user-triggerable email sources. EC-C2-02, EC-C6-06, EC-C8-02.	None.
NFR-06	No lost side effects and no partial records (D-05).	Must	Per-consumer outbox rows; enqueue in the caller's transaction; idempotency keys;	Every state change and its events commit together; every retry is idempotent. EC-C3-	None.

			reaper; redrive	07, 08, EC-C8-03, 04.	
NFR-07	Runs within a student budget, about $7 per month (D0 Part 6, D1 §6).	Should	One Render service, Supabase free tier, in-process pollers and scheduler, no broker	No component adds a hosted service except E1 (free tier).	E1 provider not chosen; free-tier quota unconfirmed (GQ-05).

4.2 Reverse check: design element to requirement

Every component C1 to C8 traces to a story (D0 Part 3). The check below covers what D1 and D2 added beyond D0, which is where scope creep would appear.

Design element	Requirement behind it	Verdict
G1 Auth middleware	US-01, NFR-03	Keep.
G2 Configuration (phrase list, holidays, keys)	NFR-04, AC-02.2, NFR-03	Keep: each file feeds a requirement.
G3 AuditLog: LISTING_RELEASED/REJECTED, APPROVAL_DECIDED	US-05, US-03 (decision must be recorded), NFR-04	Keep.
G3 AuditLog: ACCOUNT_VERIFIED	No requirement asks for it	Weak trace. Kept because one insert per sign-in is cheap and supports incident review; team to confirm (PR-S12).
C8 Suppression list and WebhookReceipt	D0 I13 (BOUNCED marks recipient undeliverable), NFR-05	Keep.
C8 priority column	US-01 (a sign-in link should not wait behind bulk notices)	Keep: one column.
C6 alert throttle columns	NFR-05	Keep.
C4 OBFUSCATION category	US-05, NFR-04	Keep.
C3 GET /listings/{id}, GET /listings/mine	US-02 (see a listing before messaging), AC-01.3 (owner sees held status later)	Keep: added because D1 had no read endpoint (PR-S11).
C2 GET /auth/me (considered)	None: C1 reads role and expiry from the token	Cut. Not in D2.
Self-service manager sign-up (considered)	None in User_Stories (D-09 defers it)	Cut. Listed as GQ-04.
Listing withdraw endpoint (considered)	None in User_Stories	Not added; open question GQ-03.
GET /healthz	NFR-07 (host health check keeps one instance alive)	Keep.

5. Design Rationale

Entries cover D1 and D2 decisions; D0 decisions D-01 to D-12 are referenced, not repeated. "Changed by review" marks entries the peer review altered; it will be updated after class (GQ-01). Entries marked self-audit come from checking D1 against D0 before the review.

ID	Decision	Trade-off accepted	Builds on	Changed by review
R-01	We chose one outbox row per consumer (C5, C6, C7) over D1's single event row that the relay fans out, because D1's drain only reached C5 and C8 and skipped D0's I16 and I9, and one row per consumer lets each retry, fail, and be redriven on its own.	We accepted the trade-off of three rows per publish and a consumer column to maintain because volume is a few events a day and D-12 already requires that a slow consumer never blocks the others.	Corrects D1 §3 Algorithm B; builds on D-05, D-12	Self-audit (PR-S02)
R-02	We chose a 5 s outbox poll (D0 I8) over D1's 30 s poll because 30 s plus one 30 s retry already reaches 60 s and misses AC-01.1's search budget.	We accepted the trade-off of about 17,000 cheap index-only queries a day because the partial index keeps each under a millisecond and the database is otherwise idle.	Corrects D1 §3; restores D0 I8	Self-audit (PR-S01)
R-03	We chose a click-to-confirm page (POST consumes the token) over consuming the token when the emailed link is opened, because Microsoft 365 link scanning on UC Mail can open links before the student does and would burn the token.	We accepted the trade-off of one extra click in sign-in because a consumed-before-use link would make AC-01.2's 24 h window meaningless for those students.	Refines D-02, D-09	Self-audit (PR-S05)
R-04	We chose literal phrase matching on word boundaries, a substitution-folded second pass, and a mixed-script flag over a regex list or a hosted moderation API, because literals cannot backtrack (no ReDoS against the 2 s budget), results show the exact phrase (AC-01.3), and no listing text leaves our system.	We accepted the trade-off of missing creative spellings outside the substitution map because a reviewer queue that only sees deterministic, explainable flags is what US-05 can test, and a classifier can still sit behind I7 later.	Builds on D-04	None yet
R-05	We chose to split I17 into getContacts (C8: email, firstName) and getEligibility (C7: role, verified) over one lookup returning every field, because C7 never needs an email address and NFR-02 says only C8 should read one.	We accepted the trade-off of two functions to maintain in C2 because the second is about ten lines and removes a privacy leak.	Revises D0 I17; builds on D-03	Self-audit
R-06	We chose one typed enqueue(tx, NotificationRequest) for I10, I11, and I12, addressed by toUserId, over three signatures with a raw email in I10 and an untyped params object in I12, because one signature gives one idempotency rule, one suppression check, and compile-time checking of each template's params.	We accepted the trade-off that C2 passes a userId it just created instead of the address the student typed because C8 needs the userId anyway to resolve suppression and contacts.	Revises D0 I10 to I12	None yet
R-07	We chose an in-process daily reminder job guarded by a Postgres advisory lock and a conditional UPDATE over an external cron service, because exactly-once (AC-02.2) then depends only on the database, and it adds no hosted service (NFR-07).	We accepted the trade-off that a day the service is down delays reminders to the next run because a late single reminder still meets 'exactly one' and the outage is already logged.	Builds on D-01, D-11	None yet
R-08	We chose one error-code rule (400 for schema and range failures, 422 for well-formed requests a business rule refuses, 409 for state conflicts) over D0's mix of 400 and 422 for the same kind of error, because C1 maps codes to one behavior each (Part 2.2) and the shared zod schema produces 400 by construction.	We accepted the trade-off of changing codes D0 published for I3, I5, and I6 because nothing is built yet and the change is listed in Part 2.10.	Revises D0 I3, I5, I6 error cells	Self-audit (PR-S03)
R-09	We chose to keep the D1 Bearer token in sessionStorage over moving it to an httpOnly cookie, because D1's API contract and C5/C6 endpoints already assume Bearer and a cookie adds CSRF handling to every POST.	We accepted the trade-off that an XSS bug could read the token because React escapes output, the CSP allows scripts only from our origin, and the token lives at most 24 h.	Builds on D-03, D1 §5	None yet
R-10	We chose to write review and approval audit rows inside the decision's own transaction over best-effort logging after commit, because a Fair Housing or approval decision with no record of who made it fails NFR-04 and US-03.	We accepted the trade-off that an audit-table failure blocks decisions (503) because decisions are rare, retryable, and worse to lose than to delay.	Builds on D-10	None yet

6. Peer Design Review Notes

To complete in class. Parts 6.2 to 6.4 are templates: record each reviewing team's feedback as action items, apply the must-fix items to D2 before submitting, then update Part 5's last column. Part 6.1 lists items the team found itself while walking D0 box by box before the review; they are not peer feedback.

6.1 Self-audit action items (pre-review)

ID	Criterion	Issue	Raised by	Action	Owner	Priority	Due	Status
PR-S01	Feasibility	D1 outbox poll is 30 s; D0 says 5 s; 30 s misses AC-01.1.	Own team (D0 walk)	Adopt 5 s; recompute budgets (Part 2.11, R-02).	Jovan	Must-fix	Oct 14	Done in D2
PR-S02	Completeness	D1 outbox drain skips I16 (C6) and I9 (C7).	Own team	Per-consumer outbox rows (Part 2.8, Fig. 6, R-01).	Jovan	Must-fix	Oct 14	Done in D2
PR-S03	Clarity	Same validation failure is 422 in D0 (I3, I5, I6) and 400 in D1.	Own team	One error-code rule (Part 1.1, R-08); revisions in Part 2.10.	Jovan, Mohamed	Must-fix	Oct 14	Done in D2
PR-S04	Clarity	D1 cites AC-01.4 for the review 409 and AC-01.1 for invalid input; D0 maps invalid input to AC-01.4.	Own team	Traceability follows D0's mapping; note in Part 4.	Jovan	Must-fix	Oct 14	Done in D2
PR-S05	Risk	UC Mail link scanning may consume the one-time token.	Own team	Click-to-confirm page (R-03); test with a real @mail.uc.edu inbox.	Danvanth	Must-fix	Week 8	Design done; test open
PR-S06	Risk	D0 worst cases (41 s, 3.6 min) omit send timeouts and poll lag.	Own team	Recompute (Part 2.11).	Danvanth	Worth considering	Oct 14	Done in D2
PR-S07	Feasibility	No way for a listing to leave PUBLISHED (withdraw, filled, expired).	Own team	Ask instructor whether to add a story; if yes, add LISTING_WITHDRAWN event.	Jovan	Worth considering	Week 8	Open (GQ-03)
PR-S08	Feasibility	Email provider not chosen; quota, webhook signing, and idempotency support unknown.	Own team	Pick provider against EC-C8-02, 04, 06 needs.	Danvanth	Must-fix	Week 8	Open (GQ-05)
PR-S09	Completeness	senderFirstName (D0 I11) has no source field.	Own team	Collect firstName at first confirmation (Part 2.3, EC-C2-06).	Danvanth	Must-fix	Oct 14	Done in D2
PR-S10	Risk	Phrase list v1 has no content; protected classes under Ohio law not confirmed.	Own team	Draft list v1 with the reviewer; confirm categories.	Jovan	Must-fix	Week 8	Open (GQ-02)
PR-S11	Completeness	No endpoint returns one listing or an owner's own listings.	Own team	Add GET /listings/{id} and /listings/mine (Part 2.8).	Jovan	Must-fix	Oct 14	Done in D2
PR-S12	Clarity	ACCOUNT_VERIFIED audit rows have no requirement behind them.	Own team	Keep or cut at next team meeting (Part 4.2).	Mohamed	Out of scope for now	Week 8	Open

6.2 Feedback received

One block per reviewing team. Criterion is one of completeness, feasibility, clarity, risk. Priority is must-fix, worth considering, or out of scope for now.

Reviewed by Team ____ (items PR-Axx)

ID	Criterion	Issue (as raised)	Action	Owner	Priority	Due
PR-A01						
PR-A02						
PR-A03						

Reviewed by Team ____ (items PR-Bxx)

ID	Criterion	Issue (as raised)	Action	Owner	Priority	Due
PR-B01						
PR-B02						
PR-B03						

6.3 Summary

Item	Notes
Criterion with the most feedback	____ (count by criterion across 6.2)
Must-fix items applied before submission	____ (list IDs and the D2 part changed)

6.4 Top two issues we raised while reviewing other teams

#	Team reviewed	Issue we raised	Criterion	Why it mattered
1				
2				